

One partition function call to rule them all: Efficiently computing the equilibrium energy landscape of multiple nucleic acid strands

Abstract

Throughout physics, chemistry, machine learning and discrete mathematics, the partition function is a normalisation factor for probability assignment. As a sum over all system states, it is often, but not always, intractable. In the nearest neighbour model of DNA/RNA strand binding, the partition function is a sum of Boltzmann-weighted free energies of the exponentially large set of secondary structures—discrete base-pairing states. Dirks, Bois, Schaeffer, Winfree and Pierce [SIAM review, 2007] gave a polynomial time partition function algorithm for multiple ($\mathcal{O}(1)$) strands, generalising McCaskill’s [Biopolymers, 1990] single-stranded algorithm. Despite the intuition that partition function should somehow describe the entire state space, thus far, different approaches were required to solve various different energy-landscape prediction problems. For instance: finding the most favoured, or minimum free energy (MFE), structure(s) for multiple strands was open since 2007, and recently solved using new techniques by Shalaby and Woods [ICALP, 2025].

Ducloz, Shalaby and Woods [DNA31, 2025] asked whether there is a generalised *magnifiable* partition function algorithm, which we answer here in the affirmative. We compute partition function in time $\mathcal{O}(N^3(c-1)!)$ and space $\mathcal{O}(N^2)$, for $c = \mathcal{O}(1)$ strands and N bases, the same overhead as previous work. However, we go on to show that magnification allows for efficient decoding of the entire energy landscape, and that *a single function call* efficiently solves and improves upon a large number of problems, including:

- MFE is computable in $\mathcal{O}(N^3(c-1)!)$ time, $\mathcal{O}(N^2)$ space; a considerable improvement over the $\mathcal{O}(N^4(c-1)!)$ time, $\mathcal{O}(N^4)$ space result of Shalaby and Woods [ICALP 2025].
- The first polynomial time algorithm for the multi-stranded *density of states* problem, also in time, space $\mathcal{O}(N^3(c-1)!)$, $\mathcal{O}(N^2)$.
- The first polynomial time algorithm outputting the entire list of energy levels occupied by at least one structure (in time, space $\mathcal{O}(N^3(c-1)!)$, $\mathcal{O}(N^2)$), and the first algorithms for various other energy landscape and counting problems.

Our approach unifies the field of energy landscape prediction showing that a single, general, algorithmic framework neatly unifies a large collection of problems that practitioners care about, and that hitherto required distinct techniques with worse time bounds.

1 Introduction

The almost-optimal information encoding capabilities of DNA and RNA are harnessed by biology to encode myriad structures and complex behaviours, and by scientists and engineers to create bespoke nanostructures [37] and molecular computers [53]. Consequently, predicting the structures formed by DNA and RNA strands is of rather wide interest. Mathematical models are deep enough that theorists have found interesting mathematical structure, and in some cases have designed useful prediction algorithms for practitioners [40, 55, 23], while needing need to know very little about biology/chemistry. In the past few decades, DNA’s proven computational abilities provide further motivation: molecular programmers have designed collections of DNA strands that form nanostructures [37] and embed computation (Boolean circuits) [36, 43, 29, 30, 14, 49, 33] leading to P-hard or NP-hard prediction problems [53], even at equilibrium [15, 48, 45].

The most used and theoretically interesting model of DNA/RNA secondary structure is the nearest neighbour model. It’s history begins in statistical mechanics [56, 8, 7, 46], and predicting

structures in that model fits with the long tradition [31] of algorithmic prediction for statistical mechanics models with state spaces of exponential size in system description [22]. When algorithmically predicting statistical mechanical systems important considerations are: What kinds of combinatorial structures populate the state space? And how are probabilities assigned to those structures? What is the shape of the energy landscape, or probability distribution?

1.1 Background: the nearest neighbour model of DNA binding

Here, we study the *nearest neighbour model* of DNA/RNA binding [40, 55, 23]. The *primary structure* of a DNA strand is a word over the alphabet of *bases* $\{A, C, G, T\}$, with U instead of T for RNA. Bases bond in pairs, A-T and C-G. A *secondary structure* (Figure 1) of one or more strands is a set of base pairs.¹ The set of all secondary structures of a multiset of strands is called the ensemble, Ω , and has size exponential in the total number of bases—since each base can be bound or not (assuming a non-trivial system). Ω is a combinatorial state space, but what we care about are which states are more or less likely, or how the entire probability distribution of states looks—i.e. we assign probabilities to states via free-energy and the Boltzmann distribution.

A secondary structure S has a free energy $\Delta G(S) \in \mathbb{R}$, where more negative means more favourable [46]. $\Delta G(S)$ is linear-time computable in the number of bases in S : simply sum over the *loops* (features) defined by base pairing (Figure 1 and Eq. (3)): adjacent base pairs or stacks are favourable (more negative), but single-stranded regions confer an unfavourable (more positive) polymer-chain entropy of logarithmic cost in loop length. A key, non-local, contribution to $\Delta G(S)$ is called *rotational symmetry*, which arises when multiple indistinguishable strands (they have the same sequence) are in the same structure. If the structure has certain symmetries, it means that multiple configurations (arrange of strands) are actually the same structure hence these “extra” symmetries need to be factored out so as avoids erroneous overcounting of structures when computing probabilities. The higher the degree of symmetry $R \geq 1$ of the secondary structure the greater its free energy (by positive $O(\log R)$), which lowers its probability under the Boltzmann distribution.² Other features of the nearest neighbour model are local, in the sense they are amenable to decomposition by dynamic programming, but rotational symmetry is non-local meaning that decomposed subproblems may form part of rotationally symmetric structures or not—the subproblem itself does not encode this information leading to a history of algorithmic challenges [11, 42].

Under the Boltzmann distribution the probability of a secondary structure S at equilibrium is

$$\text{Prob}[S] = \frac{1}{Q} \cdot e^{-\Delta G(S)/k_B T} \quad \text{where} \quad Q = \sum_{S \in \Omega} e^{-\Delta G(S)/k_B T} \quad (1)$$

is a real number called the partition function used as the normalisation factor in $\text{Prob}[S]$, and where k_B, T are constants that can be ignored, or set to 1, in this work (Boltzmann’s constant and temperature in Kelvin).

¹In this paper, we are concerned only about DNA/RNA secondary structures, which are lists of base pairs with associated free energies/probabilities. We are not concerned with 3D molecular structure, nor any other chemical nor biological details.

²DNA strands with the same DNA sequence are indistinguishable in the nearest neighbour model and, for all intents and purposes, experimentally (in statistical mechanics, two identical particles are indistinguishable). Hence there can be several “configurations” (rearrangements of strands) of a secondary structure S that are indistinguishable. Specifically, if identical strands can be rearranged according to a circular permutation without changing S , there are multiple identical/indistinguishable configurations of S . In order to treat these $R > 1$ identical configurations as one secondary structure in the probability space, $\Delta G(S)$ includes a positive $O(\log R)$ term, lowering probability by factor R under the Boltzmann distribution. Hence all of these “rotationally symmetric” configurations are counted as one structure. This is formalised using cyclic permutations in Section 2.3.

Input type	Partition function	MFE	#SSEL
Single strand	$\mathcal{O}(N^3)$ [28] ⁵	$\mathcal{O}(N^3)$ [58, 57, 35, 34, 51] ⁵	$\mathcal{O}(N^3)$ [Theorem 3] ^{***}
Multiple strands, bounded, i.e. $c = \mathcal{O}(1)$ strands	$\mathcal{O}(N^3(c-1)!)$ [11] ⁵ [Theorem 1] [*]	$\mathcal{O}(N^3(c-1)!)$ [Theorem 2] ^{**}	$\mathcal{O}(N^3(c-1)!)$ [Theorem 3] ^{****}
Multiple strands, unbounded, i.e. $c = \mathcal{O}(N)$ strands	#P hard [16]	APX hard [5]	#P complete [16]

^{*} Same runtime as [11], but *magnifiable*, leads to new results: Theorems 2, 3, 34, 41 and 43

^{**} Improvement over $\mathcal{O}(N^4(c-1)!)$ in Ref. [42].

^{***} Improvement over $\mathcal{O}(N^5)$ in Ref. [9].

^{****} First polynomial time algorithm.

Table 1: Run-time bounds for algorithms computing partition function, MFE and #SSEL for unpseudoknotted⁴ nucleic acid systems, and hardness results. N is the sum of all strand lengths, or number of DNA/RNA bases. Positive results are proven in the nearest neighbour model studied here, negative/hardness results in the simpler base pair matching (BPM) model [5, 16], and all results assume unpseudoknottedness since these problems are NP-hard or #P-hard with pseudoknots [16].

1.2 The challenge of handling rotational symmetry in partition function & MFE

Since the partition function is a sum over the entire state space Ω , intuition suggests that this single real number might somehow encode information about the entire energy landscape. Is this information obfuscated or algorithmically accessible? Can one extract the MFE from the partition function? What other energy landscape information, or thermodynamic predictions, can be extracted via this intuition?

To date, however, intuition has not been fortified with proof, in the sense that somewhat distinct techniques have been used to solve partition function versus other prediction problems in the nearest neighbour model. The main blockage to a unifying framework has been rotational symmetry, which, as a global property of a secondary structure does not seem amenable to the kind of dynamic programming (DP) algorithms used to date for partition function. I.e. small pieces of a secondary structure may be part of several structures, some having and some not having, rotational symmetry. To see why this matters, we take the MFE problem as an example since, until this paper, it has required different algorithmic techniques from partition function. The partition function algorithm of Dirks, Bois, Schaeffer, Winfree and Pierce [11] uses McCaskill-style dynamic programming [28], with novel characterisation theorems, to correctly count structures so that a single, final, algebraic step can correct for rotation symmetry penalties. In other words, rotational symmetry is not handled during dynamic programming recursions. That approach, of algebraically handling rotational symmetry outside of dynamic programming did not automatically carry over to the MFE problem—since MFE uses minimisation, instead of summation, there was no obvious way to algebraically correct a multitude of candidate solutions for differing rotational symmetry inaccuracies in a single, final step. Ref. [11] left MFE open, solved recently [42] by first computing a (possibly incorrect) MFE that ignores rotational symmetry, but then proving a characterisation lemma about rotational symmetry that yields a quadratic bound on how far the algorithm needs to then backtrack [54] to find the true MFE. Unfortunately, backtracking meant expensive time and space bounds of $\mathcal{O}(N^4(c-1)!)$ and $\mathcal{O}(N^4)$ —i.e. factor- N and factor- N^2 in time and space respectively, worse than Dirks, Bois, Schaeffer, Winfree and Pierce [11].

There are many other problems one can ask about the energy landscape, one of the most all-encompassing is *density of states* (DOS, or #SSEL [16]) where we seek an efficient description of the entire energy landscape, specifically the set of energy levels that are occupied and the number of

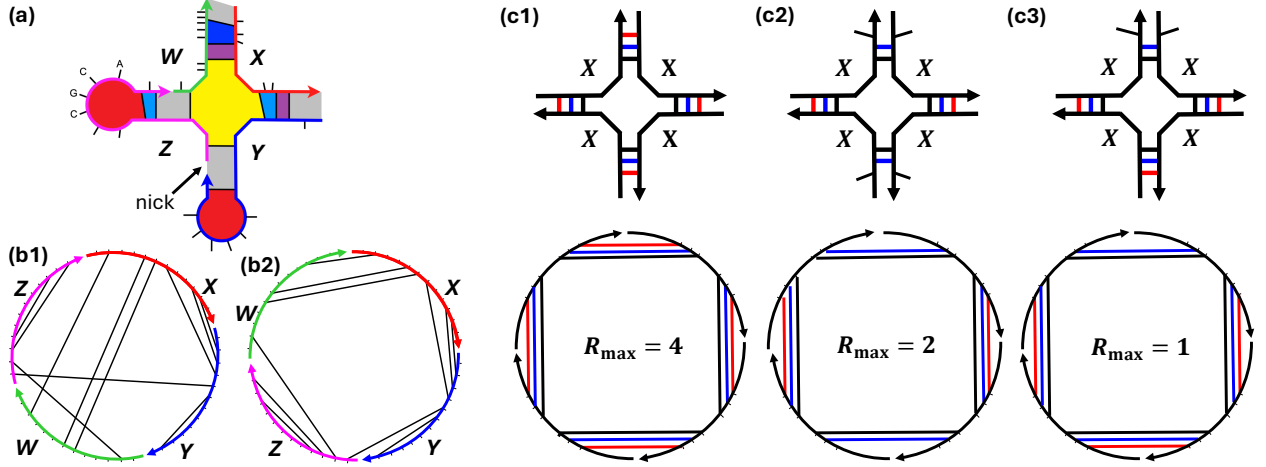


Figure 1: (a) Illustration of one of the many possible secondary structures for $c = 4$ DNA strands W, X, Y, Z . Short black ticks: DNA bases (a few are shown ... C, G, C, A...). Black lines: base pairs (drawing not to scale). Loops are colour-coded: stack (purple), multiloop (yellow), hairpin (red), bulge (light blue), internal (dark blue), external (grey). The nick separates two strands. Formally, the secondary structure is a set of base pairs: $S = \{(1_Z, 5_Y), (2_Z, 2_W) \dots, (6_Y, 12_Y)\}$. (b1) Polymer graph for strand ordering $\pi' = WZXY$, denoted $\text{Poly}(S, \pi')$, with base-pair crossings. (b2) By reordering to $\pi = ZWXY$ we get another polymer graph $\text{Poly}(S, \pi)$ for S , without crossings, hence S is unpsuedoknotted. (c1–3): Three secondary structures with their associated polymer graphs, each with different maximum symmetry degree $R_{\max}$, and each composed of the same four indistinguishable (identical; same DNA sequence) strands of type X . (c1) $R_{\max} = 4$ (rotation by 90° or 180° gives the same secondary structure). (c2) $R_{\max} = 2$ (rotation by 180° gives the same secondary structure). (c3) $R_{\max} = 1$ (asymmetric secondary structure). Each structure has 4 nests on the top, right, bottom, and left of their polymer graphs.

structures per level. DOS is especially useful for simulations of the dynamics of DNA/RNA folding, since transition rates can be derived from that description, and regions dense in state may indicate (so called) kinetic folding traps [26]. To date, there is no polynomial time algorithm for DOS, other than in the single stranded setting [9] ($O(N^5)$ time, Table 1). Any multi-stranded algorithm would need to simultaneously handle a multitude of rotational symmetry penalties over the entire state space, something previous algorithms are not suited to (and anyway, they merely solve partition function and MFE in the multi-stranded setting).

Here, we take a new approach, by having the algorithm decompose the partition function problem in terms of its rotational symmetry structure [3]. For that we need a more general partition function algorithm.

1.2.1 A magnifiable partition function algorithm

Ducloz, Shalaby and Woods [16] identified and defined *magnifiability* as a desirable property of a partition function algorithm³ (Figure 2), meaning that it computes over α -scaled states: something previous algorithms for the nearest neighbour model [11, 42] can not explicitly handle:

$$Q_\alpha = \sum_{S \in \Omega} e^{-\alpha \cdot \Delta G(S) / k_B T} \quad (2)$$

for any $\alpha \in \text{poly}(N)$, and where other variables are defined in Eq. (1). The magnified partition function Q_α generalises the usual partition function Q , since $\alpha = 1$ implies $Q_1 = Q$.

Ducloz, Shalaby and Woods [16] showed that the simplified base pair matching/stacking models *that ignore rotational symmetry* have a magnifiable partition function algorithm (which they

³Ref. [16] used the term “magnification adaptable” that we shorten here.

used to show hardness results in those simple models, Table 1). But they left open the problem of whether there is a magnifiable partition function algorithm for the most-studied and applied nearest neighbour model. The key challenge being rotational symmetry: how to efficiently handle the potentially exponentially-sized set of rotationally symmetric secondary structures under α -magnification? Rotational symmetry is a global property of a secondary structure, and previous local (subproblem) decomposition techniques fail to address this question.

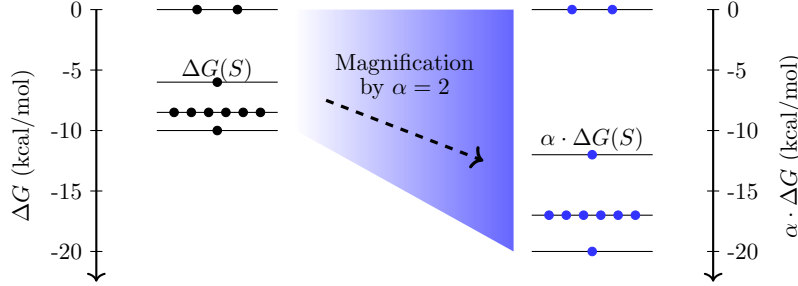


Figure 2: Magnification. Each node is a secondary structure S on an energy level, with nodes on the left having free energy $\Delta G(S)$, and right having magnified free energy $\alpha \cdot \Delta G(S)$. Magnification increases the absolute value of energy levels, without changing the distribution of secondary structures per level. From [16].

1.3 Main results

In this paper, we give a magnifiable partition function algorithm (Eq. (2)) with the same runtime as the partition function algorithm of Dirks, Bois, Schaeffer, Winfree and Pierce [11] (Eq. (1)), but that (a) explicitly handles rotational symmetry in the problem decomposition, in order to (b) unite several common prediction problems under the umbrella of partition function, which in turn will yield (c) more efficient algorithms for those problems. Our main results are:

Theorem 1. There is an $\mathcal{O}(N^3(c-1)!)$ time and $\mathcal{O}(N^2)$ space magnifiable partition function algorithm for $c = \mathcal{O}(1)$ DNA/RNA strands with N bases, in the unpsudoknotted⁴ nearest neighbour model.⁵

The proof spans Sections 3, 4, 5 and 6, with overviews in Sections 1.4 and 4.1. Setting $\alpha = 1$ gives the (usual) partition function with the same run time and space usage of Ref. [11].

With larger $\alpha \in \Theta(N)$,⁶ a single call to our magnifiable partition function algorithm solves MFE (improving on the $\mathcal{O}(N^4(c-1)!)$ time and $\mathcal{O}(N^4)$ space bounds of [42], proof in Section 7.4.1):

Theorem 2. There is an $\mathcal{O}(N^3(c-1)!)$ time and $\mathcal{O}(N^2)$ space algorithm for the Minimum Free Energy nearest neighbour unpsudoknotted secondary structure prediction problem, for $c = \mathcal{O}(1)$ DNA/RNA strands of total length N bases.

Likewise, a single call to our magnifiable partition function algorithm, with the same value for $\alpha \in \Theta(N)$ as before, yields a description of the entire energy landscape, known as density of states [9, 10, 24, 50, 16] (proof in Section 7.4.2):

⁴All results in this paper assume unpsudoknotted secondary structures, since with pseudoknots essentially all of the problems discussed here (partition function, MFE, #SSEL) are known to be NP-hard or #P-hard. See, for example, the introduction of Ref. [42] for an explanation and discussion of pseudoknots.

⁵We note that in the original literature [28, 11] that established these results, the polynomial is to the power 4 due to interior loop contributions (i.e. $\mathcal{O}(N^4)$ for single-stranded and $\mathcal{O}(N^4(c-1)!)$ for multistranded). The subsequent improvement in the exponent from 4 to 3 is achieved using techniques in Dirks and Pierce (2003) [12] to handle the energy contributions of interior loops more efficiently in dynamic programming.

⁶Results in this paper use only three values $\alpha = 0$, $\alpha = 1$, $\alpha = k_B T N / \varepsilon = \mathcal{O}(N)$ (we assume $k_B, T, \varepsilon = \mathcal{O}(1)$).

Theorem 3. There is an $\mathcal{O}(N^3(c-1)!)$ time and $\mathcal{O}(N^2)$ space algorithm for the #SSEL (density of states, or DOS) nearest neighbour unpseudoknotted secondary structure prediction problem for a set of $c = \mathcal{O}(1)$ DNA/RNA strands of total length N bases.

To further showcase the benefits of a magnifiable partition function, with a single call with $\alpha \in \Theta(N)$, and thus the same time/space bounds as before, we get various other results that further characterise various properties of the full energy landscape such as:

- The total number of secondary structures (Section 7.5 shows how this is a corollary of Theorem 3 when $\alpha \in \Theta(N)$, although one can more easily see that setting $\alpha = 0$ in Theorem 1 gives $Q_\alpha = Q_0$ in Eq. (1) which is the desired result).
- Theorem 34: the number of structures with various symmetry properties,
- Theorem 41: the number of MFE structures for a given permutation of strands,
- Theorem 43: the number of structures of a given energy level, strand permutation & symmetry degree.

1.4 Paper structure, and the key technical and conceptual ideas

We give an overview of key technical and conceptual ideas, but first the paper structure: Definitions are in Section 2. Section 3 discusses distinguishability and the mathematical relationship between the algorithmic ($\overline{\Omega}$) and physical (Ω) ensembles, or sets, of secondary structures. Section 4 summarises the strategy to prove Theorem 1 (in more detail than this section) and includes lemmas on converting from the algorithmic to physical partition function. Section 5 describes structural/combinatorial properties of secondary structures. Section 6 gives the the Master Chef and Assistant Chef algorithms, the core algorithms for Theorem 1. Section 7 provides multiple applications of our main result by essentially showing that the entire landscape (many of its properties), can be efficiently decoded from the magnified partition function value computed in Theorem 1.

1.4.1 The algorithmic and physical ensembles (sets) of secondary structures

Section 1.1 and Figure 1 describe the intuition behind rotational symmetry, and Section 1.2 gives one challenge in handling it efficiently, which we next elaborate on.

Consider a multi-stranded system of where all $c \geq 1$ strands, even if some have identical sequence, are considered to be distinct (we simply pretend they are distinct). The *algorithmic ensemble* of secondary structures denoted by $\overline{\Omega}$ contains all connected unpseudoknotted secondary structures S . Since strands are distinct, there is no rotational symmetry to consider ($R = 1$ in $\Delta G(S)$, Eq. (3)). State of the art algorithms [11, 42] use a dynamic programming paradigm that assumes (pretends) strands are distinct, by operating on the algorithmic ensemble (see Section 3 for details). However, the *physical ensemble* denoted by $\Omega \subseteq \overline{\Omega}$, is a maximal subset of distinct secondary structures S (where we consider identical-sequence strands to actually be identical/indistinguishable, in other words we count only one of their many indistinguishable structures ($1 \leq R \leq c$ in $\Delta G(S)$, Eq. (3))). The physical ensemble, Ω , is what we care about, since the statistical mechanics underlying the nearest neighbour model, and experiments, involve indistinguishable strands. In prior multi-strand work, computed algorithmic ensemble quantities were algebraically post-processed [11], or backtracked over (a form of post-processing) [42], to obtain desired physical ensemble quantities. Our partition function algorithm will decompose by distinct symmetry degrees R in order to more directly convert from the algorithmic to the physical ensemble during processing.

1.4.2 Key technical challenges for the main result: Theorem 1

To correctly handle computation of rotational symmetry penalties, we decomposed the problem into all possible rotational symmetries R : essentially computing a partition function over “slices” (sectors) of the polymer graph (Figure 1), for each relevant symmetry value of R . The goal is then to simply combine these magnified slice partition function values to get the correct global partition function. Several things can go wrong if one attempts to do this naively.

1. *The same secondary structure erroneously appears multiple times*: One immediate problem we are faced with is that the traditional McCaskill [28] recursions, also used by [11], are carefully crafted to work on full (circular) secondary structures and naive application to N/R -sized slices (sectors) gives incorrect partition function values due to erroneously overcounting secondary structures. Lemma 31 gives a combinatorial criterion telling us how to develop a new dynamic programming recursion scheme that modifies the recursions of [28, 11] to work on slices by forbidding certain base pairs, which prevents erroneous overcounting. We call this algorithm the “Master chef”, see overview in Section 6 and details in Sections 6.2 and 6.3.
2. *Slices (structures): correct handling of connectedness, α -scaling, and symmetry*: Our dynamic programming recursion on slices (segments of polymer graphs) is required to ensure a slice is connected by base pairs. Also, for each loop (Figure 1) formed by adding a base pair in a slice during dynamic programming, the correct α scaling should be applied and the the correct repetition (for R -fold symmetry). The algorithm that enforces these conditions, and others, is called the “Assistant chef”, see overview in Section 6 and details in Lemma 30 and Section 6.1.
3. *Erroneous overcounting of symmetry degrees*: For example, if a structure has maximum symmetry degree $R_{\max} = 8 \leq c$ it also has symmetry degrees of all other divisors of $R_{\max}$, i.e. $R = 4$, $R = 2$ and $R = 1$), and including all of these would lead to overestimation of the partition function. Lemma 19 uses a kind of inclusion-exclusion principle to separate these merged symmetry R classes (Eq. (24)) into disjoint maximum ($R_{\max}$) symmetry degree classes. See Section 4 for details.
4. *Running time should be no worse than [11]*: For the special case of all distinct strands ($R = 1$), our algorithm degenerates to that of [11], hence the running time is lower bounded by [11]. Our algorithm also ensures the asymptotic running time is not worse than [11], even for arbitrary (but constant) symmetry degree $R \leq c$. Secondly, to get all of the applications of our theorem, one call is sufficient (see below), which might have interest to practitioners (software packages [55, 23] typically run separate algorithms for the various problems discussed in this paper that actually have algorithms, partition function, MFE, #SSEL on single strands, etc.)

This is not intended to be a complete list of the challenges, and it includes only steps 1 and 2 of the 4-step overview (Strategy 1) in Section 4.

1.4.3 Applications: key ideas

In summary, Section 7 shows that to get all algorithmic results in the paper we need only call the magnified partition function with a single magnification parameter value $\alpha = k_B T N / \varepsilon = \Theta(N)$, and then do some simple (algebraic) post-processing to get the (usual) partition function Q in Eq. (1), and Theorems 2, 3, 34, 41 and 43.

However, it is easier to first understand what is returned by calling the magnified partition function under the various α values:

- $\alpha = 1$ (returns partition function, since $Q_\alpha = Q_1 = Q$ in Eqs. (1) and (2)).
- $\alpha = 0$ (returns the total number of distinguishable secondary structures; a trivial consequence of setting $\alpha = 0$ in Eq. (2)).

- $\alpha = k_B TN/\varepsilon = \Theta(N)$ (in this case the returned number, Q_α , contains a description of the entire energy landscape: Q_α can be efficiently decoded with no additional asymptotic time/space overhead to give Theorems 2, 3, 34, 41 and 43 and the previous two bullets).

In order to get these applications, Section 7.2 and Theorem 35 first show that in the limit as $\alpha \rightarrow \infty$, the magnified partition function is in fact the MFE—it is a property of the exponential function that the most favoured (MFE) energies get blown up enough to dominate all energies in the magnified partition function. Of course, algorithmically we desire a finite α , and Section 7.4.1 proves Theorem 2, showing that merely linear $\alpha = k_B TN/\varepsilon = \Theta(N)$ (for constant precision ε) is sufficient to extract the MFE from Q_α with a few algebraic operations.

Perhaps surprisingly, in Lemma 39 (“Hierarchical dominance of energy levels”) it is shown that the same $\alpha = k_B TN/\varepsilon = \Theta(N)$ value (and hence the same call to Q_α), can be used to extract/separate any energy level, that has structures at that level, from the set of all energy levels; which sets the scene for Section 7.4.2 which contains the proof of Theorem 3 (and Algorithm 7): first compute Q_α , then subtract the MFE value (times the number of structures with that value), then iteratively repeat the same level by level. Essentially, “peeling” each energy level off, one at a time, from lowest to highest (there are at most a polynomial number of levels). This structure on the energy landscape can also be unravelled in the opposite direction (from least to most favourable energy), by flipping the sign of α using the same procedure (see Algorithm 7).

Finally Section 7.5 gives a relatively simple algorithm showing all of the previous-mentioned results can be obtained by a single magnified partition function call.

1.5 Future work

1. Our MFE algorithm gives the MFE value but not the MFE secondary structure(s). Can the algorithm be modified to return the MFE structure with the same time/space bound? More broadly, in the context of reductions from problem A to B , where we have only a dynamic programming algorithm for B , can we use the matrices for B to construct the optimal solution of A ?
2. A single call to our partition function algorithm returns a number that can be quickly analysed to reveal the shape of the entire landscape. However, it does not encode paths through that landscape, hence: is there a polynomial time algorithm outputting an efficient description of all pathways through the energy landscape? Such a description would somehow efficiently encode an exponential size graph, where each edge is the formation or breakage of a base pair.
3. Checking all cyclic permutations of c strands yields the $(c - 1)!$ factor in the multi-stranded running times in Table 1. Boehmer, Berkemer, Will, and Ponty [3] reduce that to 3^c , but for the simple BPM model⁹. Can we make such improvements in the nearest neighbour model?
4. Can we increase the number of strands c beyond $O(1)$ and still have a polynomial time partition function or MFE algorithm? We know “not by much”, since for $c = \Theta(N)$ partition function is #P-hard [16] and MFE is NP-complete [5] in the simple BPM model.

2 Definition of multi-stranded DNA systems

Formally, a DNA strand⁷ s is a word over the alphabet of DNA *bases* $\{A, T, G, C\}$ (RNA uses U instead of T), indexed from 1 to $|s|$, where $|s|$ denotes the length of s . A base pair⁸ is a tuple

⁷Chemically, a DNA strand s is a sequence of nucleotide bases connected by *covalent bonds* which together make up the backbone of s , the left end of s corresponds to the 5' end of s and the right end corresponding to the 3' end. When drawing s we label the 3' end with an arrow which also shows the strand directionality, see Figure 1.

⁸Base pairs are *hydrogen bonds* formed between Watson-Crick pairs, namely C–G and A–T.

(i, j) such that $i < j$. For any c strands, we assign to each of them a unique distinct identifier in $\{1, \dots, c\}$. Each base is specified by a strand identifier and a position on that strand; i_s denotes the base of index i of strand s . In this paper we consider multi-stranded systems $\mathcal{S}$, i.e. a multiset of either DNA or RNA strands, and $N \in \{1, 2, 3, \dots\}$, called the system *length*, denotes the sum of the number of bases in all strands. Throughout this paper all logarithms are to the base e, meaning $\log = \ln$.

2.1 Connected unpseudoknotted secondary structures and polymer graphs

Definition 4 (Secondary structure S). For any multi-stranded system $\mathcal{S}$ of c interacting strands, a secondary structure S is a set of base pairs such that each base appears in at most one pair, i.e. if $(i_n, j_m) \in S$ and $(k_q, l_r) \in S$ then i_n, j_m, k_q, l_r are all distinct.

The *graph representation of a secondary structure* S is the graph $G = (V, E)$, where V is the set of bases of each strand $s \in \{1, \dots, c\}$, and $E = E_v \cup E_b$, where E_v is the set of *covalent backbone bonds* connecting base i_s with base $(i+1)_s$ for all bases $i = 1, 2, \dots, |s|-1$ on all strands $s \in \{1, \dots, c\}$, and $E_b = S$ is the set of base pairs in S .

The set of circular permutations, Π , of c strands has size $(c-1)!$ [4] (e.g., for the three strands $\{A, B, C\}$, $\Pi = \{ABC, ACB\}$), since the orderings ABC , BCA , and CAB are the same on a circle. Next, we define a polymer graph for each ordering $\pi \in \Pi$, see also Figure 1.

Definition 5 (Polymer graph). For any secondary structure S , and any strand ordering π of its c strands, the polymer graph representation of S , denoted $\text{Poly}(S, \pi)$, is a graph representation of S , embedded in the unit disk from $\mathbb{R}^2$, where the c strands are placed in succession from their 5' to 3' ends around the circumference of the circle in the order given by π , the bases, V , are spaced evenly around the circle circumference, each element of E_v is represented by an arc on the circumference between covalently-bonded bases, and each element of E_b is represented by a chord between two different bases. See Figure 1.

Definition 6 (Unpseudoknotted secondary structure). A secondary structure S is unpseudoknotted if there exists at least one circular permutation $\pi \in \Pi$ such that $\text{Poly}(S, \pi)$ is planar (none of the chords cross), otherwise S is pseudoknotted. An example is shown in Figure 1.

A secondary structure S is connected if the graph representation of S is a connected graph. In this work, we only consider **connected unpseudoknotted** secondary structures of multi-stranded systems $\mathcal{S}$ with $c = \mathcal{O}(1)$ strands and N bases, i.e. c is a constant independent of N which is considered to be the input size. And following recent literature [57, 28, 12, 13, 11, 23], our algorithms are analysed on the unit cost RAM model [6].

2.2 The nearest neighbour energy model

Simple temperature-independent (or more accurately, temperature-ignorant) energy models for nucleic acid secondary structures, called the Base Pair Matching (BPM) model, and the Base Pair Stacking (BPS) model were proposed in literature, were used to prove hardness results [5, 16, 25].⁹ Beyond temperature dependence, simple models do not account for a number of features of DNA

⁹In the BPM model, the free energy of a secondary structure S is simply the number of base pairs in S , and in BPS it is the number of stacks in S . Although these models are not an accurate reflection of the underlying statistical mechanics since they ignore multiple important features that matter for prediction (temperature, single-stranded loops, rotational symmetry), hardness results in such simple settings intuitively suggest hardness should carry over to the more realistic nearest neighbour model.

and RNA that provide significant free energy contributions: single-stranded regions and rotational symmetry. This motivates the most realistic energy model in physics which is called the nearest neighbour energy model for any secondary structure $S \in \Omega$, where Ω is the set of all connected unpsudoknotted secondary structures of multi-stranded systems $\mathcal{S}$.

Definition 7 (The Nearest Neighbour (NN) energy model). Let S be an unpsudoknotted connected secondary structure over a multi-stranded system $\mathcal{S}$ of $c \geq 1$ strands, and where strands with the same sequence are treated as indistinguishable. S is decomposed into loops, each of one the types: hairpin, interior, exterior, stack, bulge, or multiloop, as described in Figure 1. Then, the free energy of any secondary structure $S \in \Omega$, $\Delta G(S, T)$ or simply $\Delta G(S)$, when temperature T is clear in the context, is defined as the sum of the following three terms:

$$\Delta G(S) = \sum_{l \in S} \Delta G(l) + (c - 1)\Delta G^{\text{assoc}} + k_B T \log R_{\max}(S) \quad (3)$$

- The first is the sum of the free energies $\Delta G(l)$ of S 's constituent loops [11], where the free energy for each loop $\Delta G(l)$ is defined with respect to the free energy of the unpaired reference state [40].
- ΔG^{assoc} is the helix association [11] penalty applied for each of the $c - 1$ strands added to the first strand to form a complex of c strands to guarantee the connectedness of S .
- $R_{\max}(S)$ is the maximum degree of rotational symmetry of S , defined in Section 2.3. In particular, since favourable free energies are usually negative, the term $k_B T \log R_{\max}(S) \geq 0$ corresponds to the reduction in the entropic contribution of S , as any $R_{\max}$ -fold rotationally symmetric secondary structure has a corresponding $R_{\max}$ -fold reduction in its distinguishable conformational space [11], see Figure 1 and Remark 12, in Section 2.3.

2.3 Multi-stranded rotational symmetry

Intuitively, in Section 2.1 we assigned each one of the c strands a unique identifier, dealing with them as *distinct* strands even if two or more have the same sequence. However, in statistical mechanics, strands with the same sequence are *indistinguishable* in the sense that they behave identically with respect to relevant measurable quantities [11]. Mathematically, we say that *two strands are indistinguishable* if they have the same sequence. Also, *two secondary structures are indistinguishable* if their polymer graphs can be rotated so that all strands are mapped onto indistinguishable strands, all base pairs are mapped onto base pairs, and all unpaired bases are mapped onto unpaired bases; otherwise the structures are *distinct* [11, 55].

For any c strands, not necessarily distinct, they consist of $k \leq c$ *strand types*, usually denoted by uppercase English letters $X, Y, \dots$ ¹⁰ A *multi-stranded DNA system* $\mathcal{S} = \{(t_1, n_1), (t_2, n_2), \dots, (t_k, n_k)\}$, is a *multiset* of k strand types $t_1, \dots, t_k$ with repetition numbers $n_1, \dots, n_k \in \mathbb{N}$ such that $n_1 + \dots + n_k = c$.¹¹ For such a multiset $\mathcal{S}$ we can think of each circular permutation π as a string over strand types such that each strand type t_i appears exactly n_i times (e.g., $\mathcal{S} = \{(X, 6), (Z, 3)\}$ one valid π is $XZXXZXZX$).

Definition 8 (Symmetry degree of a permutation [42]). For any circular permutation π , we say $n \in \mathbb{N}$ is a symmetry degree of π if $\pi = y^n$ for some y , a prefix of π .

¹⁰In contrast with some of the literature. we exclude using $\{A, T, G, C\}$ for strand types; to avoid any confusion between strand and base types.

¹¹It is known [41] how to efficiently reduce the circular permutation space by getting rid of circular permutations that are redundant due to indistinguishable strands, which is important to consider when computing the partition function to avoid overcounting, but not needed for MFE.

Example 9. The set of symmetry degrees of permutation $\pi = XZXZXXZ$ is $\{1, 2, 4\}$, since $\pi = (XZXZXXZ)^1 = (XZXZ)^2 = (XZ)^4$.

For any circular permutation π , its maximum symmetry degree is denoted $v(\pi)$, and the corresponding repeating prefix x , such that $x^{v(\pi)} = \pi$, is the *fundamental component* of π . It can be seen that x is the smallest prefix that repeats over π . Indeed, $v(\pi)$ is the number of cyclic permutations that map each strand to a strand of the same type. Any repeating prefix of π must be a multiple of its fundamental component [42].

Remark 10 (Notation: X_m^n). For any circular permutation π , its *augmented version* gives full ordering information for each fundamental component. For example, if $\pi = XYXZXYXZ$, then its fundamental component is $XYXZ$, and $X_1^1Y_1^1X_2^1Z_1^1 X_1^2Y_1^2X_2^2Z_1^2$ is augmented version of π , such that X_m^n , means the m th strand of type X in the n th fundamental component of π .

We can visualize any ordering π by representing it as a *regular* $v(\pi)$ -gon with each of its $v(\pi)$ vertices representing a fundamental component. Let $\rho = (1\ 2\ 3\ \dots\ v(\pi))$,¹² and consider the cyclic group¹³ G^π generated by ρ . Intuitively, G^π is the group of all $v(\pi)$ rotational motions in plane of the regular $v(\pi)$ -gon that give the same $v(\pi)$ -gon. We can represent G^π as follows: $G^\pi = \{\rho^0, \rho^1, \dots, \rho^{v(\pi)-1}\}$, where ρ^i represents rotation of the regular $v(\pi)$ -gon by the angle of $i \times \frac{360^\circ}{v(\pi)}$, where $|G^\pi| = v(\pi)$.

Intuitively, the rotational symmetry of a secondary structure S with strand ordering π is the number of rotations of its polymer graph $\text{Poly}(S, \pi)$ that give the same polymer graph, as shown in Figure 1.

Definition 11 (R -fold rotationally symmetric structure [42]). A connected unpseudoknotted secondary structure S and strand ordering π (and thus polymer graph $\text{Poly}(S, \pi)$) is R -fold rotationally symmetric, if for any base pair (i, j) in the polymer graph $\text{Poly}(S, \pi)$ the rotation of that base pair by multiples of $(360^\circ/R)$ is also in $\text{Poly}(S, \pi)$. More formally: $(i_{X_k^l}, j_{Y_m^n}) \in \text{Poly}(S, \pi)$, iff $(i_{X_k^{a(l)}}, j_{Y_m^{a(n)}}) \in \text{Poly}(S, \pi)$ for all $a \in H \leq G^\pi$, where H is any subgroup¹⁴ satisfying the condition, and if $|H| = R$.

Remark 12 (R vs $R_{\max}$). In Definition 11, a symmetric secondary structure S can have more than one degree of symmetry. For example, if S is 4-fold rotationally symmetric, S is also 2-fold rotationally symmetric. The highest degree of symmetry that S can obtain is denoted by $R_{\max}(S)$, see Definition 7. In general, any R -fold rotationally symmetric secondary structure is also R' -fold rotationally symmetric if R' divides R , denoted by $R'|R$. As a consequence any secondary structure can be considered a 1-fold rotationally symmetric structure (or in other words, asymmetric).

2.4 PF, MFE, #SSEL problems, and magnification

Now, we define the main computational problems for which we are going to establish algorithmic relationships in the nearest neighbour model, through the lens of energy model magnification concept.

Definition 13 (Partition Function: Q).

Input: Multi-stranded system $\mathcal{S}$ with $c \geq 1$ strands, N bases, and a temperature $T \geq 0$.

Output: The partition function $Q = \sum_{S \in \Omega} e^{-\Delta G(S)/k_B T}$.

¹²Here we use algebraic cycle notation. The order of ρ , denoted by $o(\rho)$, is the length of ρ which is $v(\pi)$.

¹³ G^π is isomorphic to $C_{v(\pi)}$, cyclic group of order $v(\pi)$.

¹⁴ $H \leq G$, notationally means H is a subgroup of G . Every subgroup of cyclic group is also cyclic.

Definition 14 (Minimum Free Energy: MFE).

Input: Multi-stranded system $\mathcal{S}$ with $c \geq 1$ strands, N bases, and a temperature $T \geq 0$.

Output: The minimum free energy $\text{MFE} = \min \left\{ \Delta G(S) \mid S \in \Omega \right\}$.

Definition 15 (Density of states: #SSEL).

Input: Multi-stranded system $\mathcal{S}$ with $c \geq 1$ strands, N bases, and a temperature $T \geq 0$, and an energy value $L \in \mathbb{R}$.

Output: #SSEL(L): number of structures $S \in \Omega$ such that $\Delta G(S) = L$.

Note 16 (Energy model magnification [16]). For a positive real number $\alpha \in \mathbb{R}^+$ the notation $\alpha \cdot \Delta G(S)$ simply means to multiply the free energy $\Delta G(S)$ by α . Whenever we apply magnification in this paper, we do so uniformly over all secondary structures $S \in \Omega$, which does not change their relative free energy ordering nor distribution per energy levels, see Figure 2.

Definition 17 (Magnifiable energy model [16]). The nearest neighbour is **magnifiable** if there exists a polynomial time algorithm that for any $\alpha \in \text{poly}(N)$ computes the α -magnified partition function:

$$Q_\alpha = \sum_{S \in \Omega} e^{-\alpha \cdot \Delta G(S)/k_B T}$$

This definition captures the main question that we positively answer in this paper: *Is the nearest neighbour energy model magnifiable?*

3 Distinguishability complications: $\bar{\Omega}$ versus Ω

3.1 The algorithmic ensemble $\bar{\Omega}$ and the symmetry-naive nearest neighbour model

Consider a multi-stranded system $\mathcal{S}$ of $c \geq 1$ distinct strands (or if we pretend they are), then say we are in the *symmetry-naive nearest neighbour model*, we set $R_{\max}(S) = 1$, hence $k_B T \log R_{\max}(S) = 0$ in Eq. (3), meaning the free energy of a secondary structure S simplifies to:

$$\overline{\Delta G}(S) = \sum_{l \in S} \Delta G(l) + (c-1) \Delta G^{\text{assoc}} \quad (4)$$

Then, let $\bar{\Pi}$ be the set of all $(c-1)!$ circular permutations of the strands of $\mathcal{S}$, and let $\bar{\Omega}$ be the set (or *algorithmic ensemble*) of all connected unpsudoknotted secondary structures over $\mathcal{S}$.

For any circular permutation $\pi \in \bar{\Pi}$, let $\bar{\Omega}(\pi) \subseteq \bar{\Omega}$ be the subset of $\bar{\Omega}$ such that each connected unpsudoknotted secondary structure $S \in \bar{\Omega}(\pi)$ is representable as a crossing-free polymer graph with circular permutation π . The representation theorem of [11] (Theorem 2.1) shows that the sets $\bar{\Omega}(\pi)$, for all $\pi \in \bar{\Pi}$, form a partition of $\bar{\Omega}$; in other words, for each connected unpsudoknotted secondary structure S , there is exactly one permutation $\pi \in \bar{\Pi}$ such that $S \in \bar{\Omega}(\pi)$.

To compute the partition function $\bar{Q}$ over $\bar{\Omega}$, a known strategy is to first compute the partition function for each of the $(c-1)!$ circular permutations π separately, denoted $\bar{Q}(\pi)$:

$$\bar{Q}(\pi) = \sum_{S \in \bar{\Omega}(\pi)} e^{-\overline{\Delta G}(S)/k_B T} \quad (5)$$

Then $\bar{Q}$ is simply a sum $\bar{Q} = \sum_{\pi \in \bar{\Pi}} \bar{Q}(\pi)$ since the representation theorem [11] guarantees the computed partition function considers every secondary structure $S \in \bar{\Omega}$ exactly once, more precisely

it gives the second equality:

$$\overline{Q} = \sum_{S \in \overline{\Omega}} e^{-\overline{\Delta G}(S)/k_B T} = \sum_{\pi \in \overline{\Pi}} \sum_{S \in \overline{\Omega}(\pi)} e^{-\overline{\Delta G}(S)/k_B T} = \sum_{\pi \in \overline{\Pi}} \overline{Q}(\pi) \quad (6)$$

We let snMFE , and $\text{snMFE}(\pi)$ denote the minimum free energy over $\overline{\Omega}$ and $\overline{\Omega}(\pi)$ respectively, and we call it the symmetry naive MFE:

$$\text{snMFE} = \min_{S \in \overline{\Omega}} \overline{\Delta G}(S) = \min_{\pi \in \overline{\Pi}} \left\{ \min_{S \in \overline{\Omega}(\pi)} \overline{\Delta G}(S) \right\} = \min_{\pi \in \overline{\Pi}} \{\text{snMFE}(\pi)\} \quad (7)$$

3.2 The physical ensemble Ω and the symmetry-aware nearest neighbour model

Here we consider the more realistic, and challenging, *symmetry-aware nearest neighbour model*, or simply the nearest neighbour model see Definition 7, where for a multi-stranded system $\mathcal{S}$ of c strands, we treat strands with the same DNA/RNA sequence as indistinguishable from each other, which aligns with physical and experimental realities. The energy model is that from Eq. (3), and to emphasise the relationship between the symmetry-naive (Section 3.1) and the symmetry-aware models we note: $\Delta G(S) = \overline{\Delta G}(S) + k_B T \log R_{\max}(S)$.

Let $\Omega \subseteq \overline{\Omega}$, be a maximal subset of *distinct* secondary structures. Specifically, Ω denotes the set (called the *physical ensemble*) of all *distinct* connected unpsudoknotted secondary structures over $\mathcal{S}$, and Π to denote the set of *distinct*¹⁵ circular permutations of $\mathcal{S}$. And for any circular permutation $\pi \in \Pi$, we use $\Omega(\pi) \subseteq \Omega$ to be the subset of Ω such that each connected unpsudoknotted secondary structure $S \in \Omega(\pi)$ is representable as a crossing-free polymer graph with circular permutation π . Analogously to the algorithmic ensemble, the partition function Q is:

$$Q = \sum_{S \in \Omega} e^{-\Delta G(S)/k_B T} = \sum_{\pi \in \Pi} \sum_{S \in \Omega(\pi)} e^{-\Delta G(S)/k_B T} = \sum_{\pi \in \Pi} Q(\pi) \quad (8)$$

The representation theorem [11] gives the second equality since it holds in the physical ensemble. The MFE over the physical ensemble Ω is:

$$\text{MFE} = \min_{S \in \Omega} \Delta G(S) = \min_{\pi \in \Pi} \left\{ \min_{S \in \Omega(\pi)} \Delta G(S) \right\} = \min_{\pi \in \Pi} \{\text{MFE}(\pi)\} \quad (9)$$

3.3 Why consider the algorithmic ensemble at all?

If we threw up our hands, and only used explicit, exponential time, enumeration of all secondary structures, then we would drop the algorithmic ensemble altogether and only consider the physical ensemble Ω to predict physical systems: indistinguishable strands with the same DNA/RNA sequences are considered to be physically identical [1, 11]. So why do we care about the algorithmic ensemble at all? Pretending strands are distinct is a matter of algorithmic necessity [55], since all of the state-of-the-art algorithms use a fast dynamic programming paradigm that operates on the algorithmic ensemble. Dynamic programming is not obviously applicable to the physical ensemble: the algorithms act locally on matrix entries, unaware of the globally context, meaning they treat strands as distinct even if they have the same DNA/RNA sequence.

¹⁵For example if we have three strands X , X and Y , i.e. $\mathcal{S} = \{(X, 2), (Y, 1)\}$, then $\Pi = \{XXY\}$ and $\overline{\Pi} = \{X_1X_2Y, X_2X_1Y\}$. In general, $\Pi \subseteq \overline{\Pi}$ hence $|\Pi| \leq |\overline{\Pi}|$, a fact which becomes important when computing Q and while avoiding unintended algorithmic overcounting [11].

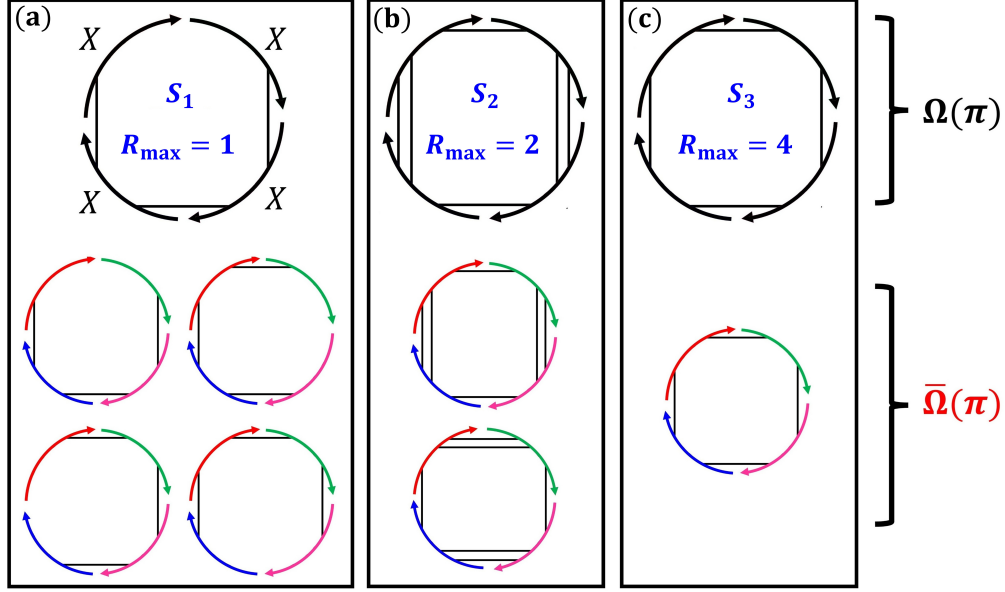


Figure 3: The symmetry-overcounting relationship between $\Omega(\pi)$ and $\bar{\Omega}(\pi)$, see Example 18

For example in the context of the partition function, since it is computed over $\bar{\Omega}$ rather than Ω . If the strand ordering can be mapped onto itself by a cyclic permutation [11], some indistinguishable secondary structures are treated as distinct by the algorithm (i.e. see Figure 3 for an example), resulting in an **overcounting** of the corresponding partition function contributions. This algorithmic approach implies that all computed physical quantities over $\bar{\Omega}$ (algorithmic ensemble) need to be post-processed [11, 42] to obtain the corresponding quantities for Ω (physical ensemble).

Specifically, for partition function, the algorithmic approach of [11] to compute $Q(\pi)$ for indistinguishable strands, is to first compute $\bar{Q}(\pi)$ (Eq. (5)), then post-process $\bar{Q}(\pi)$ by dividing by $v(\pi)$ to simultaneously cancel out overcounting and to correct the energy model from $\bar{\Delta G}(S)$ to $\Delta G(S)$ to include the entropic penalty for rotational symmetry [1], resulting in $Q(\pi)$ at the end, see Section A for more details. And for MFE, the algorithmic approach of Shalaby and Woods [42] for indistinguishable strands, is to first find the snMFE(π) (Eq. (7)), then post-process by backtracking over a polynomial set of secondary structures near the snMFE(π) to find the true MFE(π).

3.4 The algebraic relationship between $\Omega(\pi)$ and $\bar{\Omega}(\pi)$

Fully understanding the algebraic relationship between $\Omega(\pi)$ and $\bar{\Omega}(\pi)$, plays a significant role in our magnifiable partition function algorithm, since it sheds a light on what exactly should be done in our post-processing stage to compute the correct physical quantities, which is the magnifiable partition function in our case. Let's start with an example to pave the road for what we want to formalize.

Example 18. In Figure 3, since the state-of-the-art dynamic programming partition function algorithm [11] to compute $Q(\pi)$ runs over $\bar{\Omega}(\pi)$ instead $\Omega(\pi)$ (followed by a post-processing step), secondary structure $S_1 \in \Omega(\pi)$ with $R_{\max}(S_1) = 1$, will be encountered four times during the recursions, Figure 3 (a). Hence, the partition function contribution of S_1 will be overcounted by a factor of four. Comparing to structure S_2 where $R_{\max}(S_2) = 2$, which will be encountered two times and its partition function contribution overcounted by a factor of two, see Figure 3 (b). With respect

to S_3 where $R_{\max}(S_3) = 4$, its contribution will be counted only once to the partition function, see Figure 3 (c). That's one of the reasons why the algorithm of [11] requires a post-processing step, see Section A.

The main intuition we get from Example 18 is that: the lower the maximum degree of symmetry a secondary structure $S \in \Omega(\pi)$ has, the more S will be overcounted during the partition function algorithm calculation over $\overline{\Omega}(\pi)$. Now, we algebraically formalize this intuition.

Since $\Pi \subseteq \overline{\Pi}$, for any $\pi \in \Pi$, consider secondary structure $S \in \overline{\Omega}(\pi)$. A permutation $g \in G^\pi$ acts on S by relabeling strand identifiers: $g(S) = \{(i_{X_k^{a(l)}}, j_{Y_m^{a(n)}}) \mid (i_{X_k^l}, j_{Y_m^n}) \in S\}$. The stabilizer of S , denoted by $\text{Stab}(S) = \{g \in G^\pi \mid g(S) = S\}$, is the set of rotations of the polymer graph that map S onto itself. Note that $|\text{Stab}(S)| = R_{\max}(S)$.

The orbit of S , denoted by $\text{Orb}(S) = \{g(S) \mid g \in G^\pi\}$, is the subset of $\overline{\Omega}(\pi)$ corresponding to the images of S under all permutations of G^π . Note that the members of $\text{Orb}(S)$ represent indistinguishable secondary structures [11] within $\overline{\Omega}(\pi)$ from the perspective of $\Omega(\pi)$ (see Figure 3), and therefore they are represented by exactly one secondary structure in $\Omega(\pi)$. Hence the partition function contribution of secondary structure $S \in \Omega(\pi)$ is overcounted by a factor of $|\text{Orb}(S)|$ (i.e. when algorithms run over $\overline{\Omega}(\pi)$ followed by a post-processing step), since the recursive scheme treats elements of the orbit as algorithmically distinct even though they are physically indistinguishable [11]. The orbit-stabilizer theorem from group theory [17] provides the following relationship [11, 1]:

$$|\text{Orb}(S)| \cdot |\text{Stab}(S)| = |G^\pi| = v(\pi) \quad \forall S \in \overline{\Omega}(\pi) \quad (10)$$

Since $|\text{Stab}(S)| = R_{\max}(S)$, hence we get the symmetry-overcounting relationship (see Figure 3) for any secondary structure S as follows:

$$|\text{Orb}(S)| = \frac{v(\pi)}{R_{\max}(S)} \quad (11)$$

Since any secondary structure $S \in \Omega(\pi)$ may have different degrees of symmetries, see Remark 12, but a unique maximum symmetry degree $R_{\max}(S)$. Let's partition $\Omega(\pi)$ based on $R_{\max}$. Let $\Omega_{\max}(\pi, R)$ denote the set of all secondary structures $S \in \Omega(\pi)$, such that $R_{\max}(S) = R$, and let $\Omega(\pi, R)$ denote the set of all R -fold rotationally symmetric secondary structures (not necessarily the maximum), hence we get the following identities:

$$\Omega(\pi) = \bigcup_{R|v(\pi)} \Omega_{\max}(\pi, R) \quad (12)$$

$$\Omega(\pi, R) = \bigcup_{R'|v(\pi) \text{ and } R|R'} \Omega_{\max}(\pi, R') \quad (13)$$

Eqs. (12) and (13) could be extended directly to the algorithmic ensemble as follows:

$$\overline{\Omega}(\pi) = \bigcup_{R|v(\pi)} \overline{\Omega}_{\max}(\pi, R) \quad (14)$$

$$\overline{\Omega}(\pi, R) = \bigcup_{R'|v(\pi) \text{ and } R|R'} \overline{\Omega}_{\max}(\pi, R') \quad (15)$$

The relationship between $\Omega_{\max}(\pi, R)$ and $\overline{\Omega}_{\max}(\pi, R)$ could be achieved now using the symmetry-overcounting relationship in Eq. (11):

$$|\overline{\Omega}_{\max}(\pi, R)| = \frac{v(\pi)}{R} \cdot |\Omega_{\max}(\pi, R)| \quad (16)$$

4 Strategy: Computing partition function by pizza slices

Our goal is to compute $Q_\alpha(\pi)$, the magnifiable partition function, over $\Omega(\pi)$, see Definition 17.

Our algorithm will operate on symmetric “slices”, meaning a N/R size segment of a polymer graph $\text{Poly}(S, \pi)$, for any R such that $R|v(\pi)$ (in the maximum $R_{\max}$ and the non-maximum $R < R_{\max}$ sense, see Remark 12). Namely, $\Omega_{\max}(\pi, R)$, $\bar{\Omega}_{\max}(\pi, R)$, $\Omega(\pi, R)$, and $\bar{\Omega}(\pi, R)$, see Section 3.4. We denote their magnifiable partition functions by $Q_{\alpha, \max}(\pi, R)$, $\bar{Q}_{\alpha, \max}(\pi, R)$, $Q_\alpha(\pi, R)$, and $\bar{Q}_\alpha(\pi, R)$ respectively.

$$Q_{\alpha, \max}(\pi, R) = \sum_{S \in \Omega_{\max}(\pi, R)} e^{-\alpha \Delta G(S)/k_B T} \quad (17)$$

$$\bar{Q}_{\alpha, \max}(\pi, R) = \sum_{S \in \bar{\Omega}_{\max}(\pi, R)} e^{-\alpha \bar{\Delta G}(S)/k_B T} = \sum_{S \in \Omega_{\max}(\pi, R)} \frac{v(\pi)}{R} \cdot e^{-\alpha \bar{\Delta G}(S)/k_B T} \quad (18)$$

Note that, in Eq. (18), in the second equality we changed the underlying state space from $\bar{\Omega}_{\max}(\pi, R)$ to $\Omega_{\max}(\pi, R)$ but overcounted the contribution of every $S \in \Omega_{\max}(\pi, R)$ by $\frac{v(\pi)}{R}$ using the symmetry-overcounting relationship Eq. (11), to balance the contributions.

$$Q_\alpha(\pi, R) = \sum_{S \in \Omega(\pi, R)} e^{-\alpha \Delta G(S)/k_B T} = \sum_{R'|v(\pi) \text{ and } R|R'} Q_{\alpha, \max}(\pi, R') \quad \text{Using Eq. (13)} \quad (19)$$

$$\bar{Q}_\alpha(\pi, R) = \sum_{S \in \bar{\Omega}(\pi, R)} e^{-\alpha \bar{\Delta G}(S)/k_B T} = \sum_{R'|v(\pi) \text{ and } R|R'} \bar{Q}_{\alpha, \max}(\pi, R') \quad \text{Using Eq. (15)} \quad (20)$$

It's straightforward to show that:

$$Q_\alpha(\pi) = \sum_{R|v(\pi)} Q_{\alpha, \max}(\pi, R) \quad \text{Using Eq. (12)} \quad (21)$$

$$Q_{\alpha, \max}(\pi, R) = Q_\alpha(\pi, R) - \sum_{R < R' \leq v(\pi)} Q_{\alpha, \max}(\pi, R') \quad \text{Such that } R'|v(\pi) \text{ and } R|R' \quad (22)$$

$$\bar{Q}_{\alpha, \max}(\pi, R) = \bar{Q}_\alpha(\pi, R) - \sum_{R < R' \leq v(\pi)} \bar{Q}_{\alpha, \max}(\pi, R') \quad \text{Such that } R'|v(\pi) \text{ and } R|R' \quad (23)$$

4.1 Computing partition function via distinct maximum symmetry degrees: Overview and lemmas

For any multi-stranded system $\mathcal{S}$ with strand ordering π , our approach to design a magnifiable partition function algorithm for $Q_\alpha(\pi)$ is to algorithmically separate the two problems of (a) secondary structures overcounting over the algorithmic ensemble $\bar{\Omega}(\pi)$, and (b) the energy model correction from $\bar{\Delta G}(S)$ to $\Delta G(S)$ for the conversion from the algorithmic to the physical ensemble, as follows:

Strategy 1: Computing Partition Function by breaking it into separate maximum symmetry degrees classes (Proof overview of Theorem 1)

1. For all R such that $R|v(\pi)$, compute $\overline{Q}_\alpha(\pi, R)$, the magnified partition function over the algorithmic ensemble $\overline{\Omega}(\pi, R)$, the set of all secondary structures with symmetry degree R (not necessarily the maximum R) in the **algorithmic** ensemble. Strategy 2 has an overview and Section 6 achieves this task.
2. Use the computed $\overline{Q}_\alpha(\pi, R)$, for all $R|v(\pi)$, to separate and compute $\overline{Q}_{\alpha, \max}(\pi, R)$, the magnified partition function over $\overline{\Omega}_{\max}(\pi, R)$, the set of all secondary structures that have the same **maximum** symmetry degree R in the **algorithmic** ensemble. Lemma 19 and the “conversion” Algorithm 1 achieve this separation.
3. Use $\overline{Q}_{\alpha, \max}(\pi, R)$ to compute its parallel magnified partition function $Q_{\alpha, \max}(\pi, R)$ over $\Omega_{\max}(\pi, R)$, the set of all secondary structures that have the same maximum symmetry degree R in the **physical** ensemble. Lemma 20 and the “conversion” Algorithm 2 achieve this conversion from the algorithmic ensemble to the physical ensemble.
4. Sum the contribution of $Q_{\alpha, \max}(\pi, R)$ over every symmetry degree R such that $R|v(\pi)$ to compute the desired $Q_\alpha(\pi)$ at the end, see Eq. (21).

Lemma 19 (Algorithm $\overline{Q}_\alpha(\pi, R) \implies$ algorithm $\overline{Q}_{\alpha, \max}(\pi, R)$). Given any multi-stranded system $\mathcal{S}$ with strand ordering π , and magnification factor α . Assume the existence of a magnifiable partition function algorithm A that computes $\overline{Q}_\alpha(\pi, R)$ for any symmetry degree R such that $R|v(\pi)$, then there exists another magnifiable partition function algorithm B that computes $\overline{Q}_{\alpha, \max}(\pi, R)$. And B has the same time and space complexity of A .

Proof. Assume the existence of such an algorithm A that computes $\overline{Q}_\alpha(\pi, R)$ for any symmetry degree R such that $R|v(\pi)$. Recall Eq. (23), we claim that Algorithm 1 implements the following recursive equation for all $1 \leq R \leq v(\pi)$ such that $R|v(\pi)$:

$$\overline{Z}_{\alpha, \max}(\pi, R) = \overline{Q}_\alpha(\pi, R) - \sum_{\substack{R < R' \leq v(\pi) \\ \text{s.t. } R|R'}} \overline{Z}_{\alpha, \max}(\pi, R') \quad (24)$$

We claim that the computed $\overline{Z}_{\alpha, \max}(\pi, R) = \overline{Q}_{\alpha, \max}(\pi, R)$, and we prove its correctness by mathematical induction on R .

- **Base case:** If $R = v(\pi)$, the highest degree of symmetry, there is no $v(\pi) < R' \leq v(\pi)$, hence $\overline{Z}_{\alpha, \max}(\pi, v(\pi)) = \overline{Q}_\alpha(\pi, v(\pi))$. Since $\overline{Q}_\alpha(\pi, v(\pi)) = \overline{Q}_{\alpha, \max}(\pi, v(\pi))$, see Eq. (23), this implies that $\overline{Z}_{\alpha, \max}(\pi, v(\pi)) = \overline{Q}_{\alpha, \max}(\pi, v(\pi))$, proving the base case.
- **Induction Hypothesis:** For all $k < R \leq v(\pi)$ such that $R|v(\pi)$, assume $\overline{Z}_{\alpha, \max}(\pi, R) = \overline{Q}_{\alpha, \max}(\pi, R)$.
- **Induction step:** If $R = k$ such that $k|v(\pi)$. Apply the induction hypothesis on Eq. (24), by replacing $\overline{Z}_{\alpha, \max}(\pi, R')$ with $\overline{Q}_{\alpha, \max}(\pi, R')$, we get the following using Eq. (23), proving the induction step.

$$\overline{Z}_{\alpha, \max}(\pi, R) = \overline{Q}_\alpha(\pi, R) - \sum_{\substack{R < R' \leq v(\pi) \\ \text{s.t. } R|R'}} \overline{Q}_{\alpha, \max}(\pi, R') = \overline{Q}_{\alpha, \max}(\pi, R)$$

Since the set of divisors of $v(\pi)$ is of $\mathcal{O}(1)$ size (since $c = \mathcal{O}(1)$), then algorithm B computes $\overline{Q}_{\alpha, \max}(\pi, R)$ in the same time and space complexity of algorithm A. $\square$

Algorithm 1 Conversion algorithm B, that computes $\overline{Q}_{\alpha, \max}(\pi, R)$ for any symmetry degree R such that $R|v(\pi)$, over the distinguishable ensemble $\overline{\Omega}_{\max}(\pi, R)$, given a multi-stranded system $\mathcal{S}$ with strand ordering π , magnification factor α , and magnifiable algorithm A that computes $\overline{Q}_{\alpha}(\pi, R)$ for any symmetry degree R over the distinguishable ensemble $\overline{\Omega}(\pi, R)$.

```

1: for  $R \leftarrow 1 \dots v(\pi)$  and  $R|v(\pi)$  do                                 $\triangleright O(1)$  of  $R$ 's since  $v(\pi) = \mathcal{O}(1)$ 
2:    $\overline{Z}_{\alpha, \max}(\pi, R) = 0$ 
3:   Compute  $\overline{Q}_{\alpha}(\pi, R)$                                                      $\triangleright$  Using algorithm A
4: end for
5:  $\overline{Z}_{\alpha, \max}(\pi, v(\pi)) = \overline{Q}_{\alpha}(\pi, v(\pi))$                                  $\triangleright$  Base case
6: for  $R \leftarrow (v(\pi) - 1) \dots 1$  and  $R|v(\pi)$  do                         $\triangleright 1 \leq R < v(\pi)$  descending
7:    $Z = 0$ 
8:   for  $R' \leftarrow R + 1 \dots v(\pi)$  and  $R|R'$  do                             $\triangleright$  Summing  $O(1)$  of them
9:      $Z += \overline{Z}_{\alpha, \max}(\pi, R')$ 
10:  end for
11:   $\overline{Z}_{\alpha, \max}(\pi, R) = \overline{Q}_{\alpha}(\pi, R) - Z$ 
12: end for
13: Return  $\overline{Z}_{\alpha, \max}(\pi, R)$  for all  $R|v(\pi)$ 

```

Lemma 20 (Algorithm $\overline{Q}_{\alpha, \max}(\pi, R) \implies$ algorithm $Q_{\alpha, \max}(\pi, R)$). Given any multi-stranded system $\mathcal{S}$ with strand ordering π , symmetry degree R such that $R|v(\pi)$, and magnification factor α . Assume the existence of a magnifiable partition function algorithm B that computes $\overline{Q}_{\alpha, \max}(\pi, R)$, then there exists another magnifiable partition function algorithm C that computes $Q_{\alpha, \max}(\pi, R)$, such that C has the same asymptotic time and space complexity as B .

Proof. Assume the existence of algorithm B that computes $\overline{Q}_{\alpha, \max}(\pi, R)$ for any symmetry degree R such that $R|v(\pi)$. Algorithm 2 represents the magnifiable algorithm C that computes $Q_{\alpha, \max}(\pi, R)$.

Algorithm 2 Conversion algorithm C, that computes $Q_{\alpha, \max}(\pi, R)$ over the indistinguishable ensemble $\Omega_{\max}(\pi, R)$, given a multi-stranded system $\mathcal{S}$ with strand ordering π , symmetry degree R such that $R|v(\pi)$, magnification factor α , and a magnifiable algorithm B that computes $\overline{Q}_{\alpha, \max}(\pi, R)$ over the distinguishable ensemble $\overline{\Omega}_{\max}(\pi, R)$.

```

1: Let  $x = \overline{Q}_{\alpha, \max}(\pi, R)$                                  $\triangleright$  Compute Eq. (18) using algorithm B
2:  $x = x \cdot R/v(\pi)$                                            $\triangleright$  Getting rid of algorithmic overcounting
3:  $x = x \cdot e^{-\alpha \log R}$                                  $\triangleright$  Free energy model correction from  $\overline{\Delta G}(S)$  to  $\Delta G(S)$ 
4: Return  $x$ 

```

- Step 2 in Algorithm 2 gets rid of the algorithmic overcounting problem, since multiplying Eq. (18) by $R/v(\pi)$ gives the following:

$$\overline{Q}_{\alpha, \max}(\pi, R) \cdot \frac{R}{v(\pi)} = \sum_{S \in \Omega_{\max}(\pi, R)} e^{-\alpha \overline{\Delta G}(S)/k_B T} \quad (25)$$

- Step 3 in Algorithm 2, the underlying energy model correction from $\overline{\Delta G}(S)$ to $\Delta G(S)$ is achieved while applying the magnification factor α to the rotational symmetry penalty term. Multiplying

Eq. (25) by $e^{-\alpha \log R}$ gives the following:

$$\begin{aligned} \overline{Q}_{\alpha, \max}(\pi, R) \cdot \frac{R}{v(\pi)} \cdot e^{-\alpha \log R} &= \sum_{S \in \Omega_{\max}(\pi, R)} e^{-\alpha [\overline{\Delta G}(S) + k_B T \log R] / k_B T} \\ &= \sum_{S \in \Omega_{\max}(\pi, R)} e^{-\alpha \Delta G(S) / k_B T} = Q_{\alpha, \max}(\pi, R) \end{aligned}$$

Hence, these two steps of multiplication gives the desired algorithm C that has the same asymptotic time and space complexity as B . $\square$

One might ask, why not skip Step 1 in Strategy 1, and design an algorithm that computes $\overline{Q}_{\alpha, \max}(\pi, R)$ directly instead of $\overline{Q}_{\alpha}(\pi, R)$? The answer is that we still use McCaskill’s [28] recursive scheme, but as a sub-module with modifications in our algorithm, see Section 6.1 for full details. It turns out to be tricky, without sacrificing efficiency, to convince McCaskill’s recursive scheme to work strictly over $\overline{\Omega}_{\max}(\pi, R)$, the set of all structures with maximum symmetry degree R in the algorithmic ensemble. This happens since any R -fold rotationally symmetric secondary structure is also R' -fold rotationally symmetric if $R'|R$, see Remark 12. From the algorithmic perspective, the difficulty of separating $\overline{\Omega}_{\max}(\pi, R)$ classes for different values of R , forces us to work over $\overline{\Omega}(\pi, R)$ instead.

4.2 Empty pizza slicing and forming strategy

Step 1 in Strategy 1 is the only missing step, hence our goal now is to design a magnifiable partition function algorithm that computes $\overline{Q}_{\alpha}(\pi, R)$, for any symmetry degree R such that $R|v(\pi)$. To motivate our **slice-based** strategy to achieve step 1, in solving the MFE problem for multiple strands including rotational symmetries, Shalaby and Woods [42] start with a pizza $\odot$ (symmetric secondary structure) and slice it into isomorphic slices $\blacktriangle$.¹⁶

Intuitively, in this paper, our goal is the complete opposite, instead of cutting pizzas into slices, we wish to build pizzas from scratch by forming slices (segments with base pairs) and repeating them around the disk to compute the magnifiable partition function, as follows:

Strategy 2: Empty Pizza Slicing and Forming Strategy $\blacktriangle$ (Step 1 of Strategy 1)

- We start with an empty pizza $\odot$, a secondary structure with no base pairs (hence disconnected in the case of multiple strands) and slice it into R empty slices $\triangle$ for any valid symmetry degree R .
- Then we take one of these R empty slices $\triangle$, which we call the **template slice** $\blacktriangle$, and analyse all connected, unpseudoknotted slices $\blacktriangle$ induced by adding all possible combinations of base pairs within $\triangle$.
- We repeat that slice $\blacktriangle$ R times, joining each to the next by a backbone bond, to form an R -fold rotationally symmetric pizza $\odot$.

Note that, in Strategy 2 having each derived slice $\blacktriangle$ be connected is sufficient for connectedness of each derived pizza $\odot$, since [42] showed that a symmetric secondary structure $\odot$ is connected if and only if each of its isomorphic slices $\blacktriangle$ is also connected. Also, during the strategy, as combinations of structures are formed, their “partial” partition functions are computed and summed

¹⁶They go on to use this slicing strategy in a technical lemma that puts a quadratic bound on the number of structures a backtracking algorithm is required to search to find the MFE value.

into larger partial values, until the final partition function is obtained. Finally, some technical lemmas are employed to prevent unintended over- or under-counting of structures, see Section 6.2.

5 Structural properties of symmetric pizzas

From now on we assume a global indexing of all bases (over all strands of any multi-stranded system $\mathcal{S}$ given a strand ordering π) from 1 to N , and use the square brackets notation, $[i, i + 1]$, to denote the covalent bond² connecting bases i and $i + 1$, implying that both bases belong to the same strand. This notation also helps to differentiate the covalent bond notation $[i, i + 1]$ from the base pair (j, k) (hydrogen bond) notation. We define an ordering relation $\prec$ on the set of all covalent bonds E_v , such that $[i, i + 1] \prec [k, k + 1]$ if $i < k$.

For any connected secondary structure S with ordering π , there are two different routes from any base i to any base j around the backbone, or circumference, of $\text{Poly}(S, \pi)$ (clockwise or anticlockwise). We define the *length* function $l[[i, j]]$ to be the length of the shorter path, including both i and j , as follows:

$$l[[i, j]] = \min\{|i - j| + 1, N - |i - j| + 1\}$$

where $[[i, j]]$ is used to denote that shorter *segment* of length $l[[i, j]]$. If the segment $[[k, l]]$ is contained in the segment $[[i, j]]$, we denote that containment by $[[k, l]] \subseteq [[i, j]]$. And if covalent bond $[k, k + 1]$ is contained in segment $[[i, j]]$, we denote that by $[k, k + 1] \in [[i, j]]$.

5.1 Pizza Cuts

Definition 21 (Empty secondary structure $\bigcirc$). Given a multi-stranded system of strands with ordering π , the empty secondary structure, denoted $\bigcirc$, is the secondary structure with no base pairs, hence $\bigcirc = \phi$, and the polymer graph representation of $\bigcirc$ is denoted $\text{Poly}(\bigcirc, \pi)$.

Intuitively, an R -symmetric backbone cut slices an empty secondary structure $\bigcirc$ into R empty slices $\triangle$ or segments, defined via a set of R evenly spaced covalent bonds² along the circumference of $\text{Poly}(\bigcirc, \pi)$. Since the covalent bonds are evenly spaced, there is exactly one covalent bond b belonging to the segment $[[1, \frac{N}{R}]]$, and hence b and R are sufficient to define the cut [42]:

Definition 22 (R -symmetric backbone cut generated by a covalent bond). Given an empty secondary structure $\bigcirc$ with strand ordering π , and a valid symmetry degree $R \geq 2$, the R -symmetric backbone cut generated by covalent bond $b = [i_{A_m^n}, (i + 1)_{A_m^n}]$ such that $b \in [[1, \frac{N}{R}]]$, is $\bigotimes_R^b = \{[i_{A_m^{a(n)}}, (i + 1)_{A_m^{a(n)}}] : \text{for all } a \in H \leq G^\pi \text{ such that } |H| = R\}$. We call b a *symmetric backbone cut generator*. Figure 4 shows an example, and Remark 10 and Definition 11 explains the notation $A_m^{a(n)}$ which is the m^{th} instance of strand A in fundamental component number $a(n)$ where $n \in \{1, \dots, v(\pi)/R\}$.

Remark 23 (R -symmetric backbone cuts ordering). The number of all different R -symmetric backbone cuts is $\frac{N-c}{R}$, as each covalent bond $[i, i + 1] \in [[1, \frac{N}{R}]]$ can be used as a cut generator [42], excluding nicks. We can extend the ordering relation $\prec$ defined on covalent bonds to the set of all R -symmetric backbone cuts based on the order of their generator bonds, formally $\bigotimes_R^{b_1} \prec \bigotimes_R^{b_2}$, if $b_1 \prec b_2$.

Slicing an empty pizza $\bigcirc$ using an R -symmetric backbone cut $\bigotimes_R^b$ with generator $b = [i, i + 1]$ leads to R empty slices $\triangle$, we call the *first empty slice* of them a *template slice* (Intuitively the slice that just starts in the middle of the covalent bond b with base $i + 1$, see Figure 4). The concept of template slices of a cut plays a major role in our algorithm as we intuitively explained in Strategy 2.

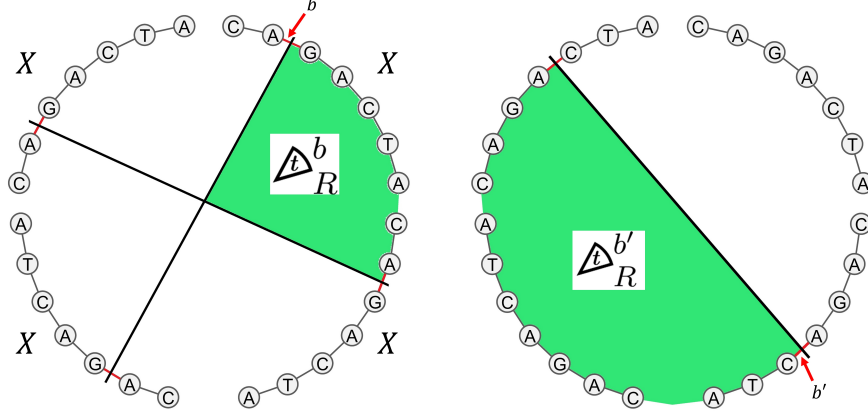


Figure 4: Empty secondary structure of multi-stranded system of 4 strands of the same type X with strand ordering $XXXX$; two different symmetric backbone cuts each associated with its template slices are shaded with green generated by b and b' respectively. (a) 4-symmetric backbone cut, which is simply the four red covalent bonds where the cut is made (b) 2-symmetric backbone cut which is the two red covalent bonds. If $b = f$, we get the prime template slice Δ_R^f .

Definition 24 (Template slice). Given an empty secondary structure $\bigcirc$ with strand ordering π , symmetry degree $R \geq 2$, and covalent bond $b = [i, i + 1] \in \llbracket 1, \frac{N}{R} \rrbracket$, then the graph G obtained from $\text{Poly}(\bigcirc, \pi)$ by removing the covalent bonds of the R -symmetric backbone cut $\bigotimes_R^b$, formally: $G = (V(\text{Poly}(\bigcirc, \pi)), E(\text{Poly}(\bigcirc, \pi)) \setminus \bigotimes_R^b)$, consists exactly of R isomorphic segments, called **empty slices**. We call the first one of these segments, the **R -template slice** denoted by Δ_R^b , more formally: For covalent bond $b = [i, i + 1]$, $\Delta_R^b = \llbracket i + 1, i + \frac{N}{R} \rrbracket$ is the R -template slice of the R -symmetric backbone cut generated by b , with base $i + 1$ (the first index of Δ_R^b) as its **first border**, and we denote it by $\mathcal{B}_1(\Delta_R^b) = i + 1$, and with base $i + \frac{N}{R}$ (the second index of Δ_R^b) as its **second border**, and we denote it by $\mathcal{B}_2(\Delta_R^b) = i + \frac{N}{R}$. Note that $\mathcal{B}_1(\Delta_R^b) < \mathcal{B}_2(\Delta_R^b)$. See Figure 4 for an example.

Given a strand ordering π , we denote the set of all R -template slices of all R -symmetric backbone cuts by $\mathcal{T}_R$, where $|\mathcal{T}_R| = \frac{N-c}{R}$, since there is a template slice per cut. For the cut generated by the first covalent bond of the system, namely the covalent bond $f = [1, 2]$, we call its corresponding R -template slice $\Delta_R^f = \llbracket 2, 1 + \frac{N}{R} \rrbracket$, the **prime template slice**. For each possible R and π there is exactly one such prime template slice. We extend the ordering relation $\prec$ to the set $\mathcal{T}_R$ of all template slices of R -symmetric backbone cuts: Formally $\Delta_R^{b_1} \prec \Delta_R^{b_2}$, if $b_1 \prec b_2$.

Definition 25 (Nest). For any connected unpsudoknotted secondary structure S with strand ordering π , we call any subset of base pairs $A \subseteq S$ a **nest** if there exists a base pair $(i, j) \in A$ such that for all base pairs $(k, l) \in A$, $\llbracket k, l \rrbracket \subseteq \llbracket i, j \rrbracket$ and there is no base pair $(m, n) \in S$ such that $\llbracket i, j \rrbracket \subset \llbracket m, n \rrbracket$. We call (i, j) the **exterior** base pair of A . And we use $\# \text{nest}(S)$ to denote the number of nests in S , see Figure 1.

Remark 26. For any $(R \geq 2)$ -fold rotationally symmetric secondary structure S , note that each nick must be part of a nest, otherwise S would be disconnected or asymmetric (a contradiction), this implies that there is a connected path (see Figure 1) of covalent bonds along the backbone (circumference) of $\text{Poly}(S, \pi)$ between any two consecutive nests

Transitioning from an empty pizza $\bigcirc$ to an actual pizza $\odot$, the following definition defines a special kind of pizza cuts, that can be thought of as slicing a pizza $\odot$ through its centre into R

slices and through R backbone bonds along the circumference, but without intersecting any formed base pair.

Definition 27 (Admissible R -symmetric backbone cut). For any connected unpseudoknotted secondary structure S with strand ordering π , and a valid symmetry degree $R \geq 2$, the R -symmetric backbone cut $\bigotimes_R^b$ generated by b is admissible, if for all covalent bonds $x \in \bigotimes_R^b$, x is not “enclosed” by any nest with exterior base pair $(i, j) \in \text{Poly}(S, \pi)$, more formally: $x \notin \llbracket i, j \rrbracket$.

Lemma 15 of [42] proves that for any $(R \geq 2)$ -fold rotationally symmetric secondary structure S , there exists at least one admissible R -symmetric backbone cut. Here, we give a stronger result based on nesting that we need to highlight in order to avoid potential overcounting that may happen during the recursions of our algorithm. The proof of Lemma 28 is in Section B.1.

Lemma 28. For any $(R \geq 2)$ -fold rotationally symmetric secondary structure S , for any two consecutive nests A and B with exteriors (i, j) and (k, l) respectively, any covalent bond $[a, a + 1]$ on the path of the circumference of $\text{Poly}(S, \pi)$ from A to B is an admissible R -symmetric backbone cut generator.

5.2 Note on Central Loops and Multiloops

Given any multi-stranded system $\mathcal{S}$ with strand ordering π , a *substructure* t is a set of base pairs only over a subsequence $[i, j]$ of $\mathcal{S}$. For a substructure t , the free energy of t is the summation of the loop energies of its constituent loops: $\Delta G(t) = \sum_{l \in t} \Delta G(l)$. Note that helix association penalty ΔG^{assoc} , and rotational symmetry penalty are not included in this definition; as they are global properties of the full secondary structure not a substructure of it.

For any $(R \geq 2)$ -fold symmetric secondary structure S , Lemma 16 of [42] showed that the graph G obtained from $\text{Poly}(S, \pi)$ by removing all covalent bonds of its admissible R -symmetric backbone cut $\bigotimes_R^b$, formally $G = (V(\text{Poly}(S, \pi)), E(\text{Poly}(S, \pi)) \setminus \bigotimes_R^b)$, is disconnected and consists exactly of R connected isomorphic substructures. Each of these substructures is simply called a *slice* of $\text{Poly}(S, \pi)$, denoted by $\blacktriangle_R^{S,b}$, and we define the loop free energy of a slice as a separate substructure as the summation of the loop energies of its constituent loops as follows:

$$\Delta G(\blacktriangle_R^{S,b}) = \sum_{l \in \blacktriangle_R^{S,b}} \Delta G(l) \quad (26)$$

Remark 29 (Central loop $\mathcal{C}(S)$). For any $(R \geq 2)$ -fold rotationally symmetric secondary structure S , lemma 20 of [42] shows the necessary existence of a unique loop in the center of $\text{Poly}(S, \pi)$ surrounded by the outer base pairs of its R slices $\blacktriangle_R^{S,b}$, called the *central loop* of S , and denoted by $\mathcal{C}(S)$. If $R_{\max}(S) > 2$, then $\mathcal{C}(S)$ is a multiloop, and if $R_{\max}(S) = 2$ then $\mathcal{C}(S)$ is either a multiloop (if each slice has more than one nest), stack, or an internal loop (if each slice has exactly one nest). Hence, the free energy of S can equivalently [42] be defined as follows:

$$\Delta G(S) = R_{\max}(S) * \Delta G(\blacktriangle_{R_{\max}(S)}^{S,b}) + \Delta G(\mathcal{C}(S)) + (c - 1)\Delta G^{\text{assoc}} + k_B T \log R_{\max}(S) \quad (27)$$

$$\overline{\Delta G}(S) = R_{\max}(S) * \Delta G(\blacktriangle_{R_{\max}(S)}^{S,b}) + \Delta G(\mathcal{C}(S)) + (c - 1)\Delta G^{\text{assoc}} \quad (28)$$

Because of the crucial importance of multiloops for our algorithm, we highlight the multiloop energy model that is used in the standard dynamic programming algorithms. The free energy of a multiloop m has the following linear form [11]:

$$\Delta G(m) = \Delta G_{\text{init}}^{\text{multi}} + n_1 \cdot \Delta G_{\text{bp}}^{\text{multi}} + n_2 \cdot \Delta G_{\text{nt}}^{\text{multi}}, \quad (29)$$

where $\Delta G_{\text{init}}^{\text{multi}}$ is called the penalty for formation of the multiloop, $\Delta G_{\text{bp}}^{\text{multi}}$ is called the penalty for each of its n_1 base pairs that border¹⁷ the interior of the multiloop, and $\Delta G_{\text{nt}}^{\text{multi}}$ is called the penalty for each of the n_2 free bases inside the multiloop. For any $(R \geq 2)$ -fold rotationally symmetric secondary structure S such that $\mathcal{C}(S)$ is a multiloop m , the contribution of the two terms $n_1 \cdot \Delta G_{\text{bp}}^{\text{multi}}$ and $n_2 \cdot \Delta G_{\text{nt}}^{\text{multi}}$ is divided equally between the R slices $\blacktriangle_R^{S,b}$ of $\text{Poly}(S, \pi)$.

6 Computing $\overline{Q}_\alpha(\pi, R)$: The final piece of our magnifiable partition function algorithm

Strategy 1, in Section 4 described our overall strategy to prove Theorem 1. Steps 2, 3 and 4 of Strategy 1 were already shown, and it only remains to prove step 1 which we do here. In this section, in formal notation, we give a magnifiable partition function algorithm that computes $\overline{Q}_\alpha(\pi, R)$, for any symmetry degree R such that $R|v(\pi)$.

Metaphorically, we can imagine the whole process of making any R -fold rotationally symmetric pizza $\odot$ as an ongoing process between two chefs:

- **The Master Chef:** The “thinking brain” behind the whole process of making all possible pizzas, and ensuring that there are no undercounting or overcounting of pizzas (secondary structures) in the process. The master chef goes through each template slice $\blacktriangle$ from the set of all template slices $\mathcal{T}_R$, and decides what toppings (base pairs) should be formed there. Section 6.3 achieves this step.
- **The Assistant Chef:** Responsible for implementing what the master chef is asking in each template slice $\blacktriangle$, but repeated R times, one per each empty slice $\triangle$. They also magnify the energy contribution of every formed loop by α . Section 6.1 achieves this step.

The master and assistant chef algorithms constitute step 1 of Strategy 1, and the proof is in the remainder of this section. Together with steps 2, 3 and 4 of Strategy 1, this gives our main theorem (restated here), with the corresponding algorithm pseudocode being Algorithm 3:

Theorem 1. There is an $\mathcal{O}(N^3(c-1)!)$ time and $\mathcal{O}(N^2)$ space magnifiable partition function algorithm for $c = \mathcal{O}(1)$ DNA/RNA strands with N bases, in the unpseudoknotted nearest neighbour model.

¹⁷Every base pair that borders the interior of a multiloop m is called a *branch* of m .

Algorithm 3 A magnifiable partition function algorithm for the nearest neighbour energy model (see Theorem 1), that takes as input: $c = \mathcal{O}(1)$ strands with total number of bases (length) N , and magnification factor $\alpha \in \text{poly}(N)$. Runs in $\mathcal{O}(N^3(c-1)!)$ time and $\mathcal{O}(N^2)$ space.

```

1: Let  $Q = 0$ 
2: for  $\pi \in \Pi$  do
3:   for  $R$  such that  $R|v(\pi)$  do
4:     - Run the Assistant Chef Algorithm 5 (Lemma 30) to pre-compute the sub-
       sequence partition functions  $Q_{i,j}, Q_{i,j}^{\text{bp}}$ , and  $Q_{i,j}^m$  that will be used by the
       Master Chef algorithm.
5:     if  $R > 1$  then
6:       - Run the Master Chef Algorithm 6 (Lemma 33) to compute the magni-
         fied partition function  $\overline{Q}_\alpha(\pi, R)$  over  $\overline{\Omega}(\pi, R)$  ensemble of all secondary
         structures with symmetry degree  $R$  (not necessarily the maximum).
7:     end if
8:   end for
9:   - Run Conversion Algorithm 1 to separate and compute the magnified partition
     function  $\overline{Q}_{\alpha, \max}(\pi, R)$  over  $\overline{\Omega}_{\max}(\pi, R)$ , the algorithmic ensemble of all secondary
     structures that have the same maximum symmetry degree  $R$ .
10:  - Run Conversion Algorithm 2 to compute the magnified partition function
     $Q_{\alpha, \max}(\pi, R)$  over  $\Omega_{\max}(\pi, R)$ , the physical ensemble of all secondary structures
    that have the same maximum symmetry degree  $R$ .
11:   $Q(\pi) = \sum_{R|v(\pi)} Q_{\alpha, \max}(\pi, R)$ .
12:   $Q = Q + Q(\pi)$ 
13: end for
14: Return  $Q$ 

```

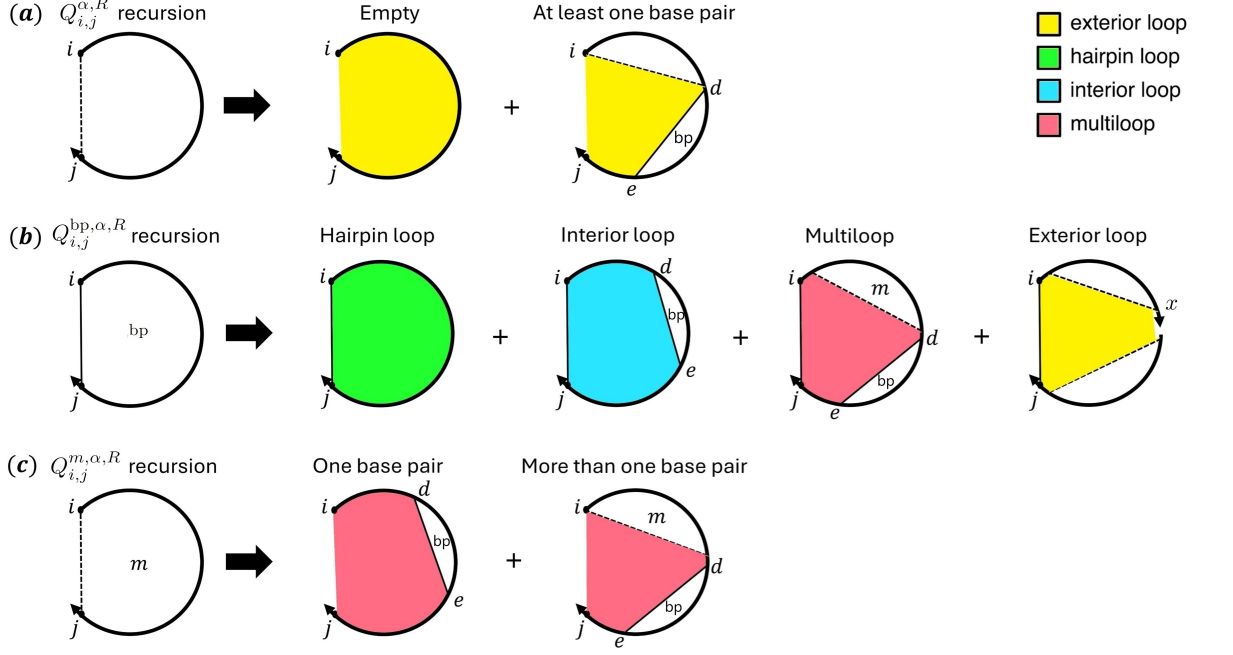
6.1 The Assistant Chef: Magnifies loop contributions by α and repeats every loop R times

In this subsection we describe the assistant chef algorithm, which is an algorithm on template slices. Specifically, the subsection contains a proof sketch of the following lemma, and the pseudocode implementation of the assistant chef algorithm, is in Section B.2:

Lemma 30 (Assistant Chef (correctness of Algorithm 5)). For any multi-stranded system $\mathcal{S}$ with strand ordering π , symmetry degree R , magnification factor α , and given any subsequence $[i, j]$ of $\mathcal{S}$, the assistant chef algorithm described in recursive Eqs. (30), (31) and (32) (Figure 5), or in Algorithm 5, correctly computes Eqs. (35), (36) and (37) in $\mathcal{O}(N^3)$ time and $\mathcal{O}(N^2)$ space.

Proof sketch. The algorithm is a straightforward generalisation the partition function recursive scheme developed by McCaskill [28] for single strands and extended by Dirks et al. [11] to multiple strands. Eqs. (30), (31) and (32), in Figure 5, represent our generalization which includes two new parameters, the repetition (or symmetry) parameter $R \in \mathbb{N}$, and the magnification parameter $\alpha \in \mathbb{Q}$. Previous work [28, 11] is the special case of $R = 1$ and $\alpha = 1$, since since that work had no repetition nor magnification.

Given any multi-stranded system $\mathcal{S}$ with strand ordering π , for any two bases i and j such that $i < j$, let $\Omega_{i,j}$ denote the set of all connected substructures over the subsequence $[i, j]$, such that it is not yet determined whether i, j are base-paired, and i, j may be on the same or different



$$Q_{i,j}^{\alpha,R} = 1 + \sum_{i \leq d < e \leq j} Q_{i,d-1}^{\alpha,R} Q_{d,e}^{bp,\alpha,R} \quad (30)$$

$$\begin{aligned}
Q_{i,j}^{bp,\alpha,R} &= \exp \{ -\alpha R \cdot \Delta G_{i,j}^{\text{hairpin}} / k_B T \} \\
&+ \sum_{i < d < e \leq j} \left[Q_{d,e}^{bp,\alpha,R} \exp \{ -\alpha R \cdot \Delta G_{i,d,e,j}^{\text{interior}} / k_B T \} \right. \\
&+ Q_{i+1,d-1}^{m,\alpha,R} Q_{d,e}^{bp,\alpha,R} \exp \left\{ -\alpha R \cdot [\Delta G_{\text{init}}^{\text{multi}} + 2\Delta G_{\text{bp}}^{\text{multi}} + (j-e-1)\Delta G_{\text{nt}}^{\text{multi}}] / k_B T \right\} \Big] \\
&+ \sum_{\substack{i \leq c < j \\ \text{s.t. } c + \frac{1}{2} \text{ is a nick}}} Q_{i+1,c}^{\alpha,R} Q_{c+1,j-1}^{\alpha,R} \quad (31)
\end{aligned}$$

$$\begin{aligned}
Q_{i,j}^{m,\alpha,R} &= \sum_{i \leq d < e \leq j} \left[Q_{d,e}^{bp,\alpha,R} \exp \left\{ -\alpha R \cdot [\Delta G_{\text{bp}}^{\text{multi}} + (d-i+j-e)\Delta G_{\text{nt}}^{\text{multi}}] / k_B T \right\} \right. \\
&\quad \left. + Q_{i+1,d-1}^{m,\alpha,R} Q_{d,e}^{bp,\alpha,R} \exp \left\{ -\alpha R \cdot [\Delta G_{\text{bp}}^{\text{multi}} + (j-e)\Delta G_{\text{nt}}^{\text{multi}}] / k_B T \right\} \right] \quad (32)
\end{aligned}$$

Figure 5: The Assistant Chef algorithm. By convention [11, 20], a solid straight line between two different bases indicates a base pair formed between them, and a dashed line demarcates a region without implying that the connected bases are paired. Shaded regions correspond to loop free energies that are explicitly incorporated at the current level of recursion. The algorithm is described by recursion diagrams (top) and recursion equations (bottom). (a) $Q_{i,j}^{\alpha,R}$ (see Eq. (30)) represents the partition function for all substructures over the subsequence $[i, j]$, magnified by α and repeated R times, such that the base pairing between i and j is undetermined. (b) $Q_{i,j}^{bp,\alpha,R}$ (see Eq. (31)) is the partition function for subsequence $[i, j]$ with the restriction that bases i and j are paired. (c) $Q_{i,j}^{m,\alpha,R}$ (see Eq. (32)) is the partition function for subsequence $[i, j]$ with the restrictions that the subsequence is inside a multiloop and contains at least one base pair. For more details see [11, 20]. Pseudocode in Algorithm 5 has some implementation details.

strands. And let $\Omega_{i,j}^{\text{bp}}$ denote the set of all connected substructures over the subsequence $[i, j]$, with the restriction that there is a base pair formed between base i and base j .

For any substructure $t \in \Omega_{i,j}$ such $|t| \geq 1$, in the case when the subsequence $[i, j]$ is supposed to be inside a multiloop m , let $\#bp(t)$ denote the number of base pairs of t that border the interior¹⁸ of m and contribute to the free energy of m , and let $\#nt(t)$ denote the number of free bases of t inside m that contribute to the free energy of m . Formally speaking, for any connected substructure $t \in \Omega_{i,j}$, $|t| \geq 1$:

$$\#bp(t) = \#nest(t), \quad (\text{as each nest is a branch}^{17} \text{ of } m \text{ (Definition 25)}) \quad (33)$$

$$\#nt(t) = (j - i + 1) - \sum_{(d,e) \in t} (e - d + 1), \quad \text{such that } (d, e) \text{ is an exterior of a nest} \quad (34)$$

For the case with no magnification nor repetition (i.e. $\alpha = R = 1$), the recursive scheme is simply that of McCaskill and Dirks et al [28, 11] which is the subsequence partition functions $Q_{i,j}^{\alpha,R}$, $Q_{i,j}^{\text{bp},\alpha,R}$, and $Q_{i,j}^{m,\alpha,R}$ in Eqs. (30), (31) and (32) in Figure 5 but with $\alpha = R = 1$. Hence if we simply remove the superscripts α and R from those recursions we get the following quantities from [28, 11]:

$$Q_{i,j} = \sum_{t \in \Omega_{i,j}} e^{-\Delta G(t)/k_B T}$$

$$Q_{i,j}^{\text{bp}} = \sum_{t \in \Omega_{i,j}^{\text{bp}}} e^{-\Delta G(t)/k_B T}$$

$$Q_{i,j}^m = \sum_{\substack{t \in \Omega_{i,j} \\ \text{s.t. } |t| \geq 1}} e^{-[\Delta G(t) + \#bp(t) \cdot \Delta G_{\text{bp}}^{\text{multi}} + \#nt(t) \cdot \Delta G_{\text{nt}}^{\text{multi}}]/k_B T}$$

The only modification in our generalized recursive equations over [28, 11] is multiplying every loop contribution¹⁹ by αR , which intuitively means that whenever the assistant chef forms a loop l in a given template slice Δ , he also magnifies its contribution by factor α and repeats it for R (all) empty slices.

Like McCaskill's [28] (and hence our) recursive scheme computes partition function by combining previously computed subsequence partition functions. By the distributive law, multiplication of subsequence partition functions (each represents a sum over a set of substructures) implicitly sums over all pairwise combinations of substructures [11]. The independence of the loop contributions in the substructureSection 5.2 energy model, $\Delta G(t) = \sum_{l \in t} \Delta G(l)$, implies that these products appropriately add the free energies in the partition function exponents. Hence multiplying each loop energy contribution, $\Delta G(l)$, by αR in Eqs. (30), (31) and (32) ends up adding their magnified and repeated version, $\alpha R \Delta G(l)$, in the exponents. As a direct consequence, the free energy of any substructure t is multiplied by αR ; since $\sum_{l \in t} \alpha R \Delta G(l) = \alpha R \sum_{l \in t} \Delta G(l) = \alpha R \Delta G(t)$.

¹⁸Note that $\#bp(t) \leq |t|$.

¹⁹Note that the 1 on the right hand side of Eq. (30) is explained as follows [28, 11]: $1 = e^{-\Delta G^{\text{ext}}/k_B T}$, where $\Delta G^{\text{ext}} = 0$ for the energy contribution of an exterior loop or equivalently in the case of having empty substructure. Hence even after magnifying the exterior loop by α and repeating it R times, their contribution to the partition function remains 1, as shown in Eq. (30).

Note that the linear form of the free energy of multiloops (Eq. (29)) is crucial here as the multiplication by αR is simply distributed over its three penalty components ($\Delta G_{\text{init}}^{\text{multi}}$, $\Delta G_{\text{bp}}^{\text{multi}}$, and $\Delta G_{\text{nt}}^{\text{multi}}$), in our generalized recursion scheme, see Eqs. (31) and (32).

Hence formally speaking given any subsequence $[i, j]$: the subsequence partition functions $Q_{i,j}^{\alpha,R}$, $Q_{i,j}^{\text{bp},\alpha,R}$, and $Q_{i,j}^{m,\alpha,R}$ in Eqs. (30), (31) and (32), for any α and R , compute the following three quantities correctly, in particular without overcounting any substructure²⁰:

$$Q_{i,j}^{\alpha,R} = \sum_{t \in \Omega_{i,j}} e^{-\alpha R \Delta G(t)/k_B T} \quad (35)$$

$$Q_{i,j}^{\text{bp},\alpha,R} = \sum_{t \in \Omega_{i,j}^{\text{bp}}} e^{-\alpha R \Delta G(t)/k_B T} \quad (36)$$

$$Q_{i,j}^{m,\alpha,R} = \sum_{\substack{t \in \Omega_{i,j} \\ \text{s.t. } |t| \geq 1}} e^{-\alpha R [\Delta G(t) + \#\text{bp}(t) \cdot \Delta G_{\text{bp}}^{\text{multi}} + \#\text{nt}(t) \cdot \Delta G_{\text{nt}}^{\text{multi}}]/k_B T} \quad (37)$$

Since the time and space complexity of McCaskill algorithm [28] is $\mathcal{O}(N^3)$ and $\mathcal{O}(N^2)$ respectively, hence also our assistant chef Algorithm 5. This completes the proof sketch of Lemma 30. $\square$

6.2 Avoiding the same pizza more than once

With respect to our empty pizza slicing and forming Strategy 2, our goal established in Section 4, of finding a magnifiable partition function algorithm that computes $\overline{Q}_\alpha(\pi, R)$ has one technical problem. Since Lemma 28 shows that any R -fold rotationally symmetric secondary structure $\odot$ may (or may not) have more than one admissible R -symmetric backbone cut. The main problem of our Strategy 2 is the possibility of considering the same secondary structure $\odot$ more than once through two recursive trajectories of the DP starting with two different template slices Δ_R^b and $\Delta_R^{b'}$. This form of algorithmic overcounting leads to compute an incorrect and overcounted partition function $\overline{Q}_\alpha(\pi, R)$.

Now, our goal is to reach a criterion that avoids that form of overcounting, and fully separates the classes of secondary structures that could be formed starting from any two different R -template slices Δ_R^b and $\Delta_R^{b'}$. For the master chef to achieve such goal, let $\mathcal{P} : \mathcal{T}_R \times E_h \rightarrow \overline{\Omega}(\pi, R)$ be a function mapping an R -template slice Δ_R^b (for any valid R and b) with exactly one single base pair (i, j) that will be added to it (i.e. $\mathcal{B}_1(\Delta_R^b) \leq i < j \leq \mathcal{B}_2(\Delta_R^b)$, and hence a possibly disconnected structure) to the set of all valid R -fold rotationally symmetric secondary structures S that have that base pair, $(i, j) \in S$, and share the admissible R -symmetric backbone cut $\otimes_R^b$. If $S \in \mathcal{P}(\Delta_R^b, (i, j))$ for some b, R and base pair (i, j) , we say S is *producible* from $(\Delta_R^b, (i, j))$. Lemma 31 establishes such criterion.

Lemma 31 (Criterion: Nest crosses the prime template slice's second border $\mathcal{B}_2(\Delta_R^f)$). Let R be the symmetry degree, hence $\Delta_R^f = \llbracket 2, 1 + \frac{N}{R} \rrbracket$ is the prime template slice. Let b and $b' \in \llbracket 2, \frac{N}{R} \rrbracket$ to be two covalent bonds such that $b \prec b'$. We claim the following:

²⁰Note that, the connectedness of any substructure t is crucial for correctness. For the simplicity of our generalized recursion scheme, the connectedness is only established within the pseudocode of Algorithm 5, we follow the approach of Dirks et al. [11] approach with respect to connectedness.

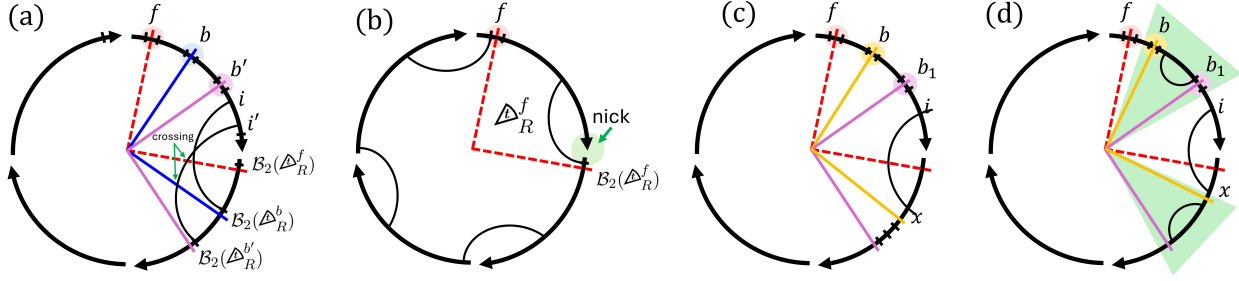


Figure 6: Crossing the the prime template slice's second border $\mathcal{B}_2(\Delta_R^f)$ criterion. Dashed red lines indicates the prime template slice borders Δ_R^f .

- First, $\mathcal{P}(\Delta_R^b, (i, x)) \cap \mathcal{P}(\Delta_R^{b'}, (i', y)) = \emptyset$, where $x = \mathcal{B}_2(\Delta_R^b)$ and $y = \mathcal{B}_2(\Delta_R^{b'})$ (i.e. x and y equal the second borders of Δ_R^b and $\Delta_R^{b'}$ respectively), and $i, i' < \mathcal{B}_2(\Delta_R^f)$ (i.e. base pair (i', y) crosses the second border of the prime template slice Δ_R^f).²¹

- $$\bigcup_{\Delta_R^b \in \mathcal{T}_R} \bigcup_{(i, x)} \mathcal{P}(\Delta_R^b, (i, x)) = \overline{\Omega}(\pi, R) \quad (38)$$

where (i, x) is the exterior base pair of a nest such that $x = \mathcal{B}_2(\Delta_R^b)$ (i.e. x equals the second border of Δ_R^b), and $\mathcal{B}_1(\Delta_R^b) \leq i < \mathcal{B}_2(\Delta_R^f)$ (i.e. (i, x) crosses the second border of the prime template slice Δ_R^f if $f < b$).

Proof. • Since $b < b'$, then $f < b'$. Since the base pair (i', y) added to template slice $\Delta_R^{b'}$ crosses the second border of the prime template slice Δ_R^f , then it also crosses the second border of Δ_R^b since $\mathcal{B}_2(\Delta_R^f) \leq \mathcal{B}_2(\Delta_R^b)$, see Figure 6 (a). For any secondary structures $S \in \mathcal{P}(\Delta_R^b, (i, x))$, $(i', y) \notin S$ or equivalently $S \notin \mathcal{P}(\Delta_R^{b'}, (i', y))$, otherwise it contradicts the fact that S has the admissible R -symmetric backbone cut $\otimes_R^b$ by the definition of $\mathcal{P}$, proving the first claim.

- First direction ($\Rightarrow$): If a secondary structure S belongs to the left hand side of Eq. (38) (i.e. S is an element of the range of $\mathcal{P}$), then $S \in \overline{\Omega}(\pi, R)$ by the definition of $\mathcal{P}$.

Second direction ($\Leftarrow$): Suppose $S \in \overline{\Omega}(\pi, R)$. If S has the admissible R -symmetric backbone cut $\otimes_R^f$ with generator $f = [1, 2]$, we are done, since we have $b = f$, and by the definition of f , there must be nest with exterior base pair (i, x) where $x = \mathcal{B}_2(\Delta_R^f) = 1 + \frac{N}{R}$ (the second border of the prime template slice Δ_R^f), otherwise S is disconnected since there is a nick between bases $\frac{N}{R}$ and $1 + \frac{N}{R} = \mathcal{B}_2(\Delta_R^f)$ (see Remark 26). See Figure 6 (b). Hence $S \in \mathcal{P}(\Delta_R^f, (i, x))$.

Otherwise, since S is an R -fold rotationally symmetric secondary structure, S may have more than one admissible R -symmetric backbone cut (see Lemma 28), $\otimes_R^{b_1}, \otimes_R^{b_2}, \dots, \otimes_R^{b_k}$, such that $f < b_1 < b_2 < \dots < b_k$. We claim that S is producible from $\Delta_R^{b_1}$, with an exterior base pair (i, x) of some nest where $i < \mathcal{B}_2(\Delta_R^f)$ and $x = \mathcal{B}_2(\Delta_R^{b_1})$ (the second border of $\Delta_R^{b_1}$).

By the definition of $\mathcal{P}$, all secondary structures S' that are producible from the prime template slice Δ_R^f are all of those that do not have a base pair crossing its own second border $\mathcal{B}_2(\Delta_R^f)$.

²¹Note that x and y are constants and only i and i' are variables here. Also, note there must be a nick between bases $\frac{N}{R}$ and $1 + \frac{N}{R} = \mathcal{B}_2(\Delta_R^f)$, see Figure 6 (b), and Remark 26, that's why in our crossing criterion $i, i' < \mathcal{B}_2(\Delta_R^f)$ to ensure contentedness, since if $i, i' = \mathcal{B}_2(\Delta_R^f)$ they are still crossing the prime template second border but introducing disconnectedness of the formed secondary structure S due to the existence of a nick.

Hence, there is a nest with an exterior base pair (i, x) added to $\Delta_R^{b_1}$ crosses the second border of Δ_R^f , in other words $i < \mathcal{B}_2(\Delta_R^f)$. We have two cases:

1. If $x = \mathcal{B}_2(\Delta_R^{b_1})$, we are done since we have satisfied the requirement of $\mathcal{P}$ in Eq. (38).
2. Otherwise $x < \mathcal{B}_2(\Delta_R^{b_1})$, there are two cases:
 - (a) All bases d in S , such that $x < d \leq \mathcal{B}_2(\Delta_R^{b_1})$ are unbound bases. Hence by the R -fold rotational symmetry of S , there is another cut generator $b \prec b_1$, where $b = [x - \frac{N}{R}, x - \frac{N}{R} + 1]$, and hence template slice Δ_R^b that satisfies the requirement of $\mathcal{P}$ in Eq. (38) such that $S \in \mathcal{P}(\Delta_R^b, (i, x))$, contradicting the fact that b_1 is the least possible R -admissible cut generator of S . See Figure 6 (c).
 - (b) There is another nest with an exterior (i', y) such $x < i' < y \leq \mathcal{B}_2(\Delta_R^{b_1})$, otherwise we violate the fact that (i, x) being an exterior base pair of nest in S . Again by the R -fold rotational symmetry of S , there is another cut generator $b \prec b_1$, where $b = [x - \frac{N}{R}, x - \frac{N}{R} + 1]$, and hence template slice Δ_R^b that satisfies the requirement of $\mathcal{P}$ in Eq. (38) such that $S \in \mathcal{P}(\Delta_R^b, (i, x))$, contradicting the fact that b_1 is the least possible R -admissible cut generator of S . See Figure 6 (d).

□

Note 32 ($\mathcal{P}(\Delta_R^b, (i, x))$ partitioning based on the number of nests). For any template slice Δ_R^b and base pair (i, j) such that $\mathcal{B}_1(\Delta_R^b) \leq i < j \leq \mathcal{B}_2(\Delta_R^b)$, the set of secondary structures $\mathcal{P}(\Delta_R^b, (i, j))$ can be logically partitioned into two disjoint sets based on the number of nests in each slice $\Delta_R^{S,b}$ of a secondary structure $S \in \mathcal{P}(\Delta_R^b, (i, j))$. (1) The first denoted by $\mathcal{P}(\Delta_R^b, (i, j), \#nest = 1)$ is the set of all secondary structures such that each has exactly one nest per slice $\Delta_R^{S,b}$. (2) The second denoted by $\mathcal{P}(\Delta_R^b, (i, j), \#nest > 1)$ is the set of all secondary structures such that each has more than one nest per slice $\Delta_R^{S,b}$. Note that if S has zero nests, it's disconnected, and hence discarded.

6.3 The Master Chef

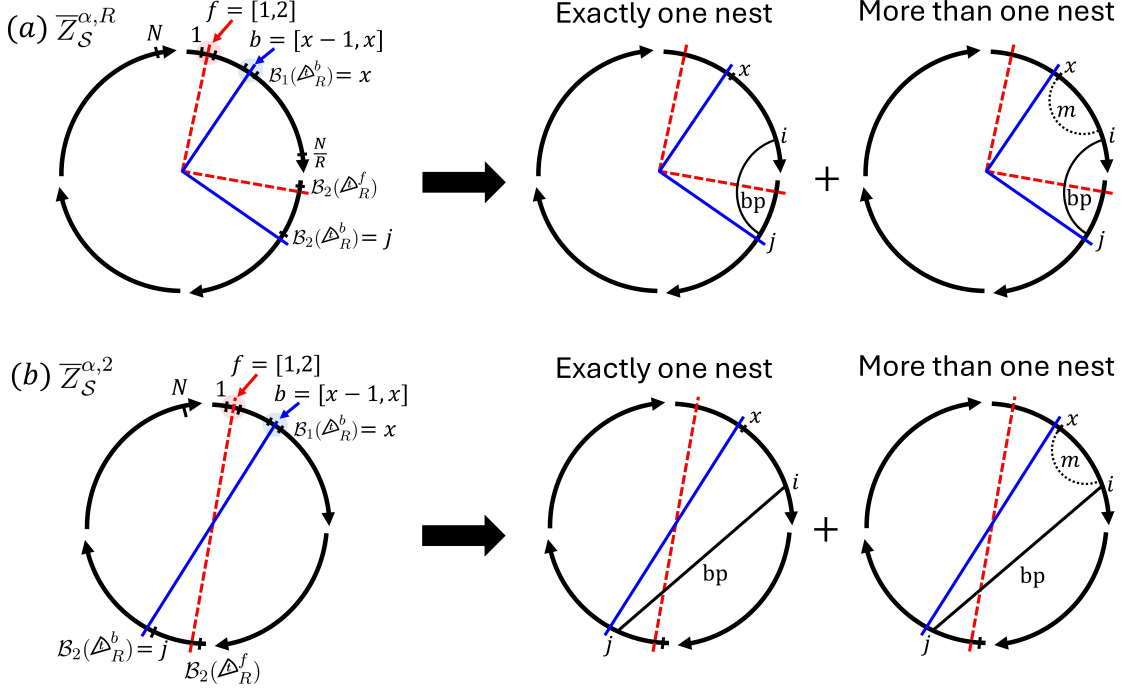
Now, we are ready to introduce our master chef algorithm, the last remaining sub-piece of our magnifiable partition function algorithm. The main job of the master chef ensures that there is no under or overcounting happening during our new proposed recursive scheme for template slices Δ . The master chef achieves this goal by incorporating our criterion of having one nest crosses the prime template slice's second border $\mathcal{B}_2(\Delta_R^f)$, see Lemma 31.

The master chef algorithm, in its new recursive scheme (Figure 7), incorporates results from Shalaby and Woods MFE algorithm [42] related to central loops of symmetric secondary structures, see Section 5.2. This allows to separate different classes of secondary structures based on the type of their central loop, and how many nests per slice each of them has. Figure 7 contains the recursion diagrams, and their corresponding recursion equations, see Eqs. (39) and (40). The proof of the following lemma and the pseudocode implementation of the master chef Algorithm 6 are in Section B.2.

Lemma 33 (Master Chef). Given any multi-stranded system of strands $\mathcal{S}$ with strand ordering π , magnification factor α , and valid symmetry degree $R \geq 2$ such that $R|v(\pi)$:

- if $R > 2$, Eq. (39) computes $\overline{Q}_\alpha(\pi, R)$ correctly. Formally speaking:

$$\overline{Z}_S^{\alpha, R} = \sum_{S \in \overline{\Omega}(\pi, R)} e^{-\alpha \overline{\Delta G}(S)/k_B T} = \overline{Q}_\alpha(\pi, R).$$



$$\begin{aligned} \overline{Z}_S^{\alpha,R} = \sum_{\Delta_R^b \in \mathcal{T}_R} \sum_{(i,j)} & \left[\mathbb{1}_{x,i} Q_{i,j}^{\text{bp},\alpha,R} e^{-\alpha [\Delta G_{\text{init}}^{\text{multi}} + R \cdot \Delta G_{\text{bp}}^{\text{multi}} + R \cdot (i-x) \Delta G_{\text{nt}}^{\text{multi}} + \beta]} / k_B T \right. \\ & \left. + Q_{x,i-1}^{m,\alpha,R} \mathbb{1}_i Q_{i,j}^{\text{bp},\alpha,R} e^{-\alpha [\Delta G_{\text{init}}^{\text{multi}} + R \cdot \Delta G_{\text{bp}}^{\text{multi}} + \beta]} / k_B T \right] \end{aligned} \quad (39)$$

$$\begin{aligned} \overline{Z}_S^{\alpha,2} = \sum_{\Delta_2^b \in \mathcal{T}_2} \sum_{(i,j)} & \left[\mathbb{1}_{x,i} Q_{i,j}^{\text{bp},\alpha,2} e^{-\alpha [\Delta G_{i,j,i',y}^{\text{interior}} + \beta]} / k_B T \right. \\ & \left. + Q_{x,i-1}^{m,\alpha,2} \mathbb{1}_i Q_{i,j}^{\text{bp},\alpha,2} e^{-\alpha [\Delta G_{\text{init}}^{\text{multi}} + 2 \Delta G_{\text{bp}}^{\text{multi}} + \beta]} / k_B T \right] \end{aligned} \quad (40)$$

Such that $\beta = (c-1)\Delta G^{\text{assoc}}$, $x = \mathcal{B}_1(\Delta_R^b)$, $j = \mathcal{B}_2(\Delta_R^b)$, $i' = i + \frac{N}{2}$, and $y = (j + \frac{N}{2}) \bmod N$. Base i of base pair (i, j) ranges²¹ as follows: $\mathcal{B}_1(\Delta_R^b) \leq i < \mathcal{B}_2(\Delta_R^b)$. And let the following two indicator variables $\mathbb{1}_{a,b}$ and $\mathbb{1}_a$ to be defined as follow:

$$\mathbb{1}_{a,b} = \begin{cases} 1 & \text{if the interval } [a, b] \text{ contains zero nicks, (i.e. hence connected)} \\ 0 & \text{otherwise} \end{cases} \quad (41)$$

$$\mathbb{1}_a = \begin{cases} 1 & \text{if } [a-1, a] \text{ is a covalent bond, (i.e. not a nick)} \\ 0 & \text{otherwise} \end{cases} \quad (42)$$

Figure 7: The Master Chef algorithm, described by recursion diagrams (top) and recursion equations (bottom), for computing the partition function for multi-stranded system $\mathcal{S}$ of $c = O(1)$ strands, given a strand ordering π , symmetry degree R , and magnification factor α . Dashed red lines indicates the prime template slice Δ_R^f borders. By convention [11, 20], a solid straight line between two different bases indicates a base pair formed between them, and a dashed line demarcates a region without implying that the connected bases are paired. (a) When $R > 2$, (b) when $R = 2$. Pseudocode implementation details are provided in Algorithm 6.

- if $R = 2$, Eq. (40) computes $\overline{Q}_\alpha(\pi, 2)$ correctly. Formally speaking:

$$\overline{Z}_S^{\alpha, 2} = \sum_{S \in \overline{\Omega}(\pi, 2)} e^{-\alpha \overline{\Delta G}(S)/k_B T} = \overline{Q}_\alpha(\pi, 2).$$

Furthermore, Eqs. (39) and (40) are implemented in the master chef Algorithm 6, that has asymptotic complexity of $\mathcal{O}(N^2)$ time, and $\mathcal{O}(1)$ space.

7 Peeling onions: Applications of the magnifiable partition function algorithm

Beside computing the partition function (i.e. set $\alpha = 1$), in this section we explore wide range of applications of our magnifiable partition function algorithm in the nearest neighbour model, from both theoretical and practical perspectives. These applications range from purely combinatorial and counting applications to the ability to efficiently compute the equilibrium landscape using just a single call to our algorithm. In Section 7.1, we prove that our magnifiable PF algorithm can be used to answer the fundamental counting questions [32] about the algorithmic and the physical ensembles $\overline{\Omega}$, Ω respectively. In Section 7.4, we prove the two remaining main results of this work, Theorem 2 and Theorem 3.

7.1 Combinatorial applications ($\alpha = 0$)

Theorem 34. Given any multi-stranded system $\mathcal{S}$ with strand ordering π , and symmetry degree R such that $R|v(\pi)$. A single call to our magnification adaptable partition function Algorithm 3, with $\alpha = 0$, can be used to compute all the following quantities correctly, in $\mathcal{O}(N^3)$ time and $\mathcal{O}(N^2)$ space:

- $|\Omega(\pi)|$, the number of secondary structures in the physical ensemble (i.e. distinguishable secondary structures).
- $|\Omega_{\max}(\pi, R)|$, the number of secondary structures with maximum degree of symmetry R such that $R|v(\pi)$ in the physical ensemble.
- $|\Omega(\pi, R)|$, the number of secondary structures with symmetry degree R , not necessarily the maximum, such that $R|v(\pi)$ in the physical ensemble.

The corresponding quantities $|\overline{\Omega}(\pi)|$, $|\overline{\Omega}_{\max}(\pi, R)|$, and $|\overline{\Omega}(\pi, R)|$ in the algorithmic ensemble could be computed also within the same call.

Proof. The intuition here is simple: we call our magnifiable partition function algorithm with magnification factor $\alpha = 0$. Hence, the energy contribution of each secondary structure $S \in \Omega(\pi)$ becomes 0, and, since $e^0 = 1$, our partition function algorithm becomes a counting algorithm, such that $Q_0(\pi) = |\Omega(\pi)|$.

Since our algorithm is slice-based, it computes $Q_{\alpha, \max}(\pi, R)$ for every $R|v(\pi)$, as a sub-module within the algorithm, see step 8 in Algorithm 3, all we need is just extract such information. Hence $Q_{0, \max}(\pi, R) = |\Omega_{\max}(\pi, R)|$. Using Eqs. (13), (14), (15) and (16) to compute $|\Omega(\pi, R)|$, $|\overline{\Omega}(\pi)|$, $|\overline{\Omega}(\pi, R)|$, and $|\overline{\Omega}_{\max}(\pi, R)|$ respectively. $\square$

7.2 Taking a limit: as $\alpha \rightarrow \infty$, the magnified partition function $Q_\alpha \rightarrow \text{MFE}$

The next theorem states that when the parameter $\alpha \rightarrow \infty$, the contribution of MFE structures dominate the α -magnified partition function. Later in Section 7.4 we will use a bounded version

of this limit to extract both the MFE value, and the number of MFE structures, from a single partition function call.

Theorem 35. Given any multi-stranded system $\mathcal{S}$ with strand ordering π :

- $\lim_{\alpha \rightarrow \infty} \frac{-k_B T \log Q_\alpha(\pi)}{\alpha} = \text{MFE}(\pi)$, i.e. the MFE value, and (43)

- $\lim_{\alpha \rightarrow \infty} \frac{Q_\alpha(\pi)}{e^{-\alpha \cdot \text{MFE}(\pi)/k_B T}} = \#\text{SSEL}(\text{MFE}(\pi))$, i.e. the number of MFE structures. (44)

Proof. Let $S^* \in \Omega(\pi)$ denote any structure with minimum free energy, i.e. $\Delta G(S^*) = \text{MFE}(\pi)$. We define a difference function $\delta : \Omega(\pi) \rightarrow \mathbb{R}_{\geq 0}$ so that for any $S \in \Omega(\pi)$, $\Delta G(S) = \text{MFE}(\pi) + \delta(S)$, where $\delta(S) \geq 0$. Hence $Q_\alpha(\pi)$ can be expressed as follows:

$$Q_\alpha(\pi) = \sum_{S \in \Omega(\pi)} e^{-\alpha [\text{MFE}(\pi) + \delta(S)]/k_B T} = e^{-\alpha \cdot \text{MFE}(\pi)/k_B T} \sum_{S \in \Omega(\pi)} e^{-\alpha \cdot \delta(S)/k_B T} \quad (45)$$

By applying the natural logarithm, and multiplying Eq. (45) by $\frac{-k_B T}{\alpha}$, we get:

$$\frac{-k_B T \log(Q_\alpha(\pi))}{\alpha} = \text{MFE}(\pi) - \frac{k_B T \log \left(\sum_{S \in \Omega(\pi)} e^{-\alpha \cdot \delta(S)/k_B T} \right)}{\alpha} \quad (46)$$

and taking the limit $\alpha \rightarrow \infty$:

$$\lim_{\alpha \rightarrow \infty} \frac{-k_B T \log(Q_\alpha(\pi))}{\alpha} = \text{MFE}(\pi) - k_B T \lim_{\alpha \rightarrow \infty} \frac{\log \left(\sum_{S \in \Omega(\pi)} e^{-\alpha \cdot \delta(S)/k_B T} \right)}{\alpha}$$

Our goal is to sandwich the right-hand side limit between lower and upper bounds that converge to 0 as $\alpha \rightarrow \infty$, yielding the first claim of the theorem statement. For the lower bound: $S^* \in \Omega(\pi)$, hence $\delta(S^*) = 0$ and $e^{-\alpha \cdot \delta(S^*)/k_B T} = 1$. The exponential function is strictly positive for all real numbers, hence $\sum_{S \in \Omega(\pi)} e^{-\alpha \cdot \delta(S)/k_B T} \geq 1$, and hence: $0 \leq \log \left(\sum_{S \in \Omega(\pi)} e^{-\alpha \cdot \delta(S)/k_B T} \right)$. For the upper bound: $\delta(S) \geq 0$, and the constants k_B and T (in Kelvin) are both positive, hence $e^{-\alpha \cdot \delta(S)/k_B T} \leq 1$ and $\sum_{S \in \Omega(\pi)} e^{-\alpha \cdot \delta(S)/k_B T} \leq |\Omega(\pi)|$, hence we get our upper bound: $\log \left(\sum_{S \in \Omega(\pi)} e^{-\alpha \cdot \delta(S)/k_B T} \right) \leq \log |\Omega(\pi)|$. Since $\lim_{\alpha \rightarrow \infty} \log |\Omega(\pi)|/\alpha = 0$, and applying the sandwich theorem [2, 38] implies that:

$$\lim_{\alpha \rightarrow \infty} \frac{\log \left(\sum_{S \in \Omega(\pi)} e^{-\alpha \cdot \delta(S)/k_B T} \right)}{\alpha} = 0, \text{ proving our first claim: } \lim_{\alpha \rightarrow \infty} \frac{-k_B T \log(Q_\alpha(\pi))}{\alpha} = \text{MFE}(\pi).$$

For our second claim, let's partition $\Omega(\pi)$ into two disjoint sets:

- The set $\mathcal{F}$ of all secondary structures S such that $\Delta G(S) = \text{MFE}(\pi)$. Note, $|\mathcal{F}| = \#\text{SSEL}(\text{MFE}(\pi))$.
- The set $\mathcal{F}'$ of all other secondary structures S such that $\Delta G(S) > \text{MFE}(\pi)$.

$Q_\alpha(\pi)$ can be written as follows:

$$Q_\alpha(\pi) = \#\text{SSEL}(\text{MFE}(\pi)) \cdot e^{-\alpha \cdot \text{MFE}(\pi)/k_B T} + \sum_{S \in \mathcal{F}'} e^{-\alpha [\text{MFE}(\pi) + \delta(S)]/k_B T}, \text{ hence:}$$

$$\frac{Q_\alpha(\pi)}{e^{-\alpha \cdot \text{MFE}(\pi)/k_B T}} = \#\text{SSEL}(\text{MFE}(\pi)) + \sum_{S \in \mathcal{F}'} e^{-\alpha \cdot \delta(S)/k_B T}. \quad (47)$$

By taking the limit, when $\alpha \rightarrow \infty$, of Eq. (47), we get the following:

$$\lim_{\alpha \rightarrow \infty} \frac{Q_\alpha(\pi)}{e^{-\alpha \cdot \text{MFE}(\pi)/k_B T}} = \#\text{SSEL}(\text{MFE}(\pi)) + \sum_{S \in \mathcal{F}'} \lim_{\alpha \rightarrow \infty} e^{-\alpha \cdot \delta(S)/k_B T}.$$

Since $|\mathcal{F}'| < |\Omega(\pi)| = \mathcal{O}(1)$, and $\delta(S) \geq 0$ by definition, then $\sum_{S \in \mathcal{F}'} \lim_{\alpha \rightarrow \infty} e^{-\alpha \cdot \delta(S)/k_B T} = 0$, proving our second claim: $\lim_{\alpha \rightarrow \infty} \frac{Q_\alpha(\pi)}{e^{-\alpha \cdot \text{MFE}(\pi)/k_B T}} = \#\text{SSEL}(\text{MFE}(\pi))$. □

7.3 Numerical Considerations

Throughout algorithmic work on the nearest neighbour model, there is seldom discussion about precision of real numbers in algorithms, perhaps for two reasons: (a) The free energies for some hardcoded features (e.g. stacks) in the nearest neighbour model are experimentally given to one or two decimal places [40, 39, 27, 21, 47], and although single-stranded loop energy scales logarithmically, precision beyond a few decimal places is somewhat meaningless experimentally. And (b) many algorithmic results have the flavour that better precision in the algorithm’s variables simply means a smooth, quantitative improvement in accuracy, i.e. no qualitative change to the result.

However, software implementations have specific precision [23, 55]. And, in this section we will assume constant, but arbitrary, precision, independent of system length (number of bases) N .

Remark 36 (Precision assumption). Throughout this work we assume the existence of a minimum gap of $\varepsilon > 0$ between any two energy levels in the nearest neighbour model. Specifically, $\varepsilon = \left(\frac{1}{10}\right)^k$ for some constant natural number $k \geq 1$, meaning k is independent of the number of bases (system size) N . In other words, for any two secondary structures $S_i, S_j \in \Omega$, $|\Delta G(S_i) - \Delta G(S_j)| = n \cdot \varepsilon$ for some $n \geq 0$.

This assumption is used in previous works [10, 23, 16, 55, 9, 24]. For example, ViennaRNA²² assumes $\varepsilon = 0.01$ kcal/mol [23]. Using this assumption, we get the following lemma:

Lemma 37. In the nearest neighbour energy model, for any multi-stranded system $\mathcal{S}$ of total length N . Let $\mathcal{G} = \{\Delta G(S) \mid S \in \Omega\}$, the set of *occupied* real valued free energies. We claim that $|\mathcal{G}| = \mathcal{O}(N)$, i.e. linear number of energy levels.

Proof. In the worst case, any secondary structure free energy is a sum of $\lfloor \frac{N}{2} \rfloor = \mathcal{O}(N)$ loop free energies. The contribution of any loop is $\mathcal{O}(1)$ using the precision assumption. This implies that for any secondary structure $S \in \Omega$, the free energy $\Delta G(S) \in \mathcal{G}$ grows linearly, i.e. $\Delta G(S) = \mathcal{O}(N)$. Hence, $|\mathcal{G}| = \mathcal{O}(N)$. $\square$

7.4 Even more efficient MFE, #SSEL, and occupied energy levels algorithms

Having these numerical considerations in mind, instead of having $\alpha \rightarrow \infty$ as in Theorem 35, the goal of Lemma 39 is to choose a finite α to achieve similar results to Theorem 35.

Note 38. For a single RNA/DNA strand, Stein and Waterman [44] showed that the asymptotic number of RNA/DNA secondary structures²³ is upper bounded by $\mathcal{U}_N = 1.104366 \cdot N^{-3/2} \cdot 2.618034^N$, where N is the number of bases. The same upper bound holds for multiple strands given a strand ordering π (in fact, a slightly lower one as multi-stranded systems require at least some base pairs to maintain connected structures). For the sake of simplicity of our analysis, we use e^N as an upper bound, since $\mathcal{U}_N < e^N$ for all $N \geq 2$. Note that, these upper bounds are system-independent, hence not tight. One of our results is an algorithm that gives the exact number of structures $|\Omega(\pi)|$ for a multi-stranded system, achieved simply by setting $\alpha = 0$, call our magnifiable partition function algorithm to get $Q_0(\pi)$, and noting that $|\Omega(\pi)| = Q_0(\pi)$, see Theorem 34. Note that $e^N > |\Omega(\pi)|$ for all N .

²²ViennaRNA [23, 18, 19, 52, 54, 9] is a widely used open-source software package designed for predicting and analyzing RNA secondary structures. It features algorithms for computing the MFE, partition function, density of states, folding kinetics, and sequence design.

²³Note that the upper bound $\mathcal{U}_N$ is with the restriction that each hairpin loop has at least one free base.

The goal of Lemma 39 is to magnify secondary structure energies so those at the any energy level L strictly dominate over all at energy levels $L' > L$, in the contribution to the α -magnified partition function, regardless of the secondary structures distribution over the energy landscape.

Lemma 39 (Hierarchical dominance of energy levels). Let $\mathcal{S}$ be a multi-stranded system of length N , strand ordering π and gaps $\geq \varepsilon = (\frac{1}{10})^k$ between energy levels (Remark 36). For any $\alpha \geq k_B T N / \varepsilon$, and for any occupied energy level L , we claim the following:

$$\sum_{S \text{ s.t. } \Delta G(S)=L} e^{-\alpha \Delta G(S)/k_B T} > \sum_{L' > L} \sum_{S' \text{ s.t. } \Delta G(S')=L'} e^{-\alpha \Delta G(S')/k_B T} \quad (48)$$

Proof. To prove our claim, we satisfy the worst case scenario of the distribution of secondary structures over the energy landscape which is as follows:

- There exist exactly one secondary structure S belonging to energy level L .
- Every other secondary structure of the system $\mathcal{S}$, belongs to the closest possible energy level L' above L , i.e. $L' = L + \varepsilon$. Even worse, we assume that e^N secondary structures belong to L' (e^N is a loose upper bound on the total number of secondary structures, see Note 38).

Assume $\alpha = k_B T N / \varepsilon$, and consider L :

$$\begin{aligned} \sum_{S \text{ s.t. } \Delta G(S)=L} e^{-\alpha \Delta G(S)/k_B T} &= \sum_{S \text{ s.t. } \Delta G(S)=L} e^{-NL/\varepsilon} \\ &\geq e^{-NL/\varepsilon} && \geq 1 \text{ structure at level } L \\ &= e^{-N(L+\varepsilon-\varepsilon)/\varepsilon} \\ &= e^N e^{-N(L+\varepsilon)/\varepsilon} \\ &> \sum_{L' > L} \sum_{S' \text{ s.t. } \Delta G(S')=L'} e^{-\alpha \Delta G(S')/k_B T} \end{aligned}$$

since $< e^N$ secondary structures have energy L' since e^N is an upper bound on the number of secondary structures, which proves our claim. $\square$

Lemma 39 will allow us to unpeel the entire energy landscape (see Theorem 3 and Algorithm 7) like an *onion*, where we can peel layers, one by one starting at the MFE level/layer and working upwards.

7.4.1 MFE

Theorem 2. There is an $\mathcal{O}(N^3(c-1)!)$ time and $\mathcal{O}(N^2)$ space algorithm for the Minimum Free Energy nearest neighbour unpseudoknotted secondary structure prediction problem, for $c = \mathcal{O}(1)$ DNA/RNA strands of total length N bases.

Proof. For any $\pi \in \Pi$, recall Eq. (46) derived in the proof of Theorem 35:

$$\frac{-k_B T \ln(Q_\alpha(\pi))}{\alpha} = \text{MFE}(\pi) - \frac{k_B T \ln\left(\sum_{S \in \Omega(\pi)} e^{-\alpha \cdot \delta(S)/k_B T}\right)}{\alpha} \quad (46)$$

Here, we call $0 \leq \theta(\alpha) = \frac{k_B T \ln\left(\sum_{S \in \Omega(\pi)} e^{-\alpha \cdot \delta(S)/k_B T}\right)}{\alpha}$, the MFE(π) approximation error. Note that $\theta(\alpha)$ is a monotonically decreasing function (a fact derived in the proof of Theorem 35 but straightforward to derive).

Our goal is to show $\theta(\alpha) < \varepsilon$ when $\alpha \geq k_B T N / \varepsilon$. In other words, the $\text{MFE}(\pi)$ approximation error is strictly less than the minimum gap between any two energy levels, and hence we can extract $\text{MFE}(\pi)$ from $Q_\alpha(\pi)$.

Assume $\alpha = k_B T N / \varepsilon$:

$$\begin{aligned}
\theta(\alpha) &= \frac{\varepsilon}{N} \log \sum_{S \in \Omega(\pi)} e^{-N \cdot \delta(S) / \varepsilon} \\
&= \frac{\varepsilon}{N} \log \left[\sum_{S \in \mathcal{F}} e^{-N \cdot \delta(S) / \varepsilon} + \sum_{S' \in \mathcal{F}'} e^{-N \cdot \delta(S') / \varepsilon} \right] && \text{partitioning of } \Omega(\pi) \text{ in Theorem 35 proof} \\
&= \frac{\varepsilon}{N} \log \left[\#\text{SSEL}(\text{MFE}(\pi)) + \sum_{S' \in \mathcal{F}'} e^{-N \cdot \delta(S') / \varepsilon} \right] && \delta(S) = 0 \\
&\leq \frac{\varepsilon}{N} \log \left[\#\text{SSEL}(\text{MFE}(\pi)) + \sum_{S' \in \mathcal{F}'} e^{-N} \right] && \delta(S') = \varepsilon, \text{ the worst candidate} \\
&< \frac{\varepsilon}{N} \log \left[\#\text{SSEL}(\text{MFE}(\pi)) + e^N e^{-N} \right] && \text{since } |\mathcal{F}'| < e^N, \text{ Note 38} \\
&< \frac{\varepsilon}{N} \log(e^N) && \text{since } \#\text{SSEL}(\text{MFE}(\pi)) + 1 < e^N \\
&= \varepsilon
\end{aligned}$$

This proves that $0 \leq \theta(\alpha) < \varepsilon$ when $\alpha = k_B T N / \varepsilon$. Hence the following equation represents an $\text{MFE}(\pi)$ prediction algorithm that runs in $\mathcal{O}(N^3)$ time and $\mathcal{O}(N^2)$ space, where rescaling by $1/\varepsilon$ and the ceiling operator remove the approximation error which we know to be $< \varepsilon$:

$$\text{MFE}(\pi) = \left\lceil \frac{-k_B T \log Q_\alpha(\pi)}{\alpha} \cdot \frac{1}{\varepsilon} \right\rceil \cdot \varepsilon \quad \text{where } \alpha \geq k_B T N / \varepsilon \quad (49)$$

Since $|\Pi| \leq (c-1)!$, Algorithm 4 predicts the MFE value in $\mathcal{O}(N^3(c-1)!)$ time and $\mathcal{O}(N^2)$ space, proving our claim. $\square$

Algorithm 4 MFE algorithm for the nearest neighbour energy model, that takes as input: $c = \mathcal{O}(1)$ strands with total number of bases (length) N . Runs in $\mathcal{O}(N^3(c-1)!)$ time and $\mathcal{O}(N^2)$ space.

```

1: Let MFE =  $+\infty$ 
2: for  $\pi \in \Pi$  do
3:   Let  $Z = Q_\alpha(\pi)$  for  $\alpha = k_B T N / \varepsilon$   $\triangleright$  Using Algorithm 3
4:    $\text{MFE}(\pi) = \left\lceil \frac{-k_B T \log Z}{\alpha} \cdot \frac{1}{\varepsilon} \right\rceil \cdot \varepsilon$ 
5:   if  $\text{MFE}(\pi) < \text{MFE}$  then
6:      $\text{MFE} = \text{MFE}(\pi)$ 
7:   end if
8: end for
9: Return MFE

```

Remark 40. The main limitation of this new efficient algorithm for MFE is that it computes the MFE value but not the corresponding MFE secondary structure.

7.4.2 #SSEL and the set of occupied energy levels

Next, for a $\pi \in \Pi$, we seek an algorithm that returns the number of secondary structures that have $\text{MFE}(\pi)$ value.

Theorem 41 (#SSEL(MFE(π)) Algorithm). Let $\mathcal{S}$ be a multi-stranded system of length N , strand ordering π and a minimum gap of $\geq \varepsilon > 0$ between energy levels (Remark 36). There is an $\mathcal{O}(N^3)$ time, $\mathcal{O}(N^2)$ space algorithm to predict #SSEL(MFE(π)), i.e. the number of MFE structures with ordering π , for the nearest neighbour model.

Proof. For any $\pi \in \Pi$, recall Eq. (47) derived in the proof of Theorem 35:

$$\frac{Q_\alpha(\pi)}{e^{-\alpha \cdot \text{MFE}(\pi)/k_B T}} = \# \text{SSEL}(\text{MFE}(\pi)) + \sum_{S \in \mathcal{F}'} e^{-\alpha \cdot \delta(S)/k_B T} \quad (47)$$

where $0 \leq \vartheta(\alpha) = \sum_{S \in \mathcal{F}'} e^{-\alpha \cdot \delta(S)/k_B T}$, is called the approximation error on the number of MFE(π) structures. Note that $\vartheta(\alpha)$ is a monotonically decreasing function.

Our goal is to show that $\vartheta(\alpha) < 1$ when $\alpha = k_B T N / \varepsilon$. In other words, the approximation error on the number of MFE(π) structures is strictly less than one, and hence we can use a simple floor function to obtain #SSEL(MFE(π)).

Assume $\alpha = k_B T N / \varepsilon$:

$$\begin{aligned} \vartheta(\alpha) &= \sum_{S \in \mathcal{F}'} e^{-N \cdot \delta(S)/\varepsilon} \\ &\leq \sum_{S \in \mathcal{F}'} e^{-N} && \delta(S) = \varepsilon, \text{ the worst candidate} \\ &< e^N \cdot e^{-N} = 1 && \text{since } |\mathcal{F}'| < e^N, \text{ Note 38} \end{aligned}$$

This proves that $0 \leq \vartheta(\alpha) < 1$ when $\alpha = k_B T N / \varepsilon$. Hence the following equation represents an algorithm, that calls our magnifiable partition function once (with a single α value) and uses a simple floor, that computes the number of MFE(π) secondary structures, i.e. #SSEL(MFE(π)), and runs in $\mathcal{O}(N^3)$ time and $\mathcal{O}(N^2)$ space.

$$\# \text{SSEL}(\text{MFE}(\pi)) = \left\lfloor \frac{Q_\alpha(\pi)}{e^{-\alpha \cdot \text{MFE}(\pi)/k_B T}} \right\rfloor, \quad \text{where } \alpha = k_B T N / \varepsilon \quad (50)$$

□

For a strand ordering $\pi \in \Pi$, in the previous proof we peeled away the MFE(π) energy level. Next, we move on and peel the next occupied energy level above MFE(π), and by keeping iterating this process we efficiently compute the full energy landscape in Algorithm 7. The correctness proof of Theorem 3 and Algorithm 7 are in Section B.3.

Theorem 3. There is an $\mathcal{O}(N^3(c-1)!)$ time and $\mathcal{O}(N^2)$ space algorithm for the #SSEL (density of states, or DOS) nearest neighbour unpseudoknotted secondary structure prediction problem for a set of $c = \mathcal{O}(1)$ DNA/RNA strands of total length N bases.

Note 42. We should note that Algorithm 7 peels energy levels in ascending order starting from the optimal MFE(π) level, but also could be adapted to run in descending order by using $\alpha = -k_B T N / \varepsilon$, where the negative sign reverses the energy landscape without affecting the relative energies nor its distribution.

7.4.3 Revealing the energy landscape at a higher resolution (rotational symmetry)

Since our magnifiable partition function algorithm has a slice-based approach, and it computes $Q_{\alpha, \max}(\pi, R)$ for every R such that $R|v(\pi)$, as a sub-module within the algorithm, see step 8 in Algorithm 3. We can again benefit from our algorithmic slice-based construction, and run the onion peeling Algorithm 7 on each of $Q_{\alpha, \max}(\pi, R)$ for all R 's separately. This leads to higher resolution information about the energy landscape of any system $\mathcal{S}$. For example, for any $\pi \in \Pi$, with a single call to our magnifiable algorithm we can answer a question like:

- How many structures $S \in \Omega(\pi)$ that belong to energy level L , such that $R_{\max}(S) = R$?

This leads to Theorem 43, the higher resolution version of Theorem 3:

Theorem 43 (Onion Peeling: High Resolution Version). Given any multi-stranded system $\mathcal{S}$ of total length N , with strand ordering π , symmetry degree R such that $R|v(\pi)$, energy level L . There exists an $\mathcal{O}(N^3)$ time, $\mathcal{O}(N^2)$ space algorithm to predict $\#SSEL(L, \pi, R)$, i.e. the number of secondary structures $S \in \Omega(\pi)$ at energy level L with maximum symmetry degree R , i.e. $\Delta G(S) = L$, and $R_{\max}(S) = R$.

7.5 One partition function call to rule them all

We showed different useful applications for our new magnifiable partition function algorithm. These applications range from (usual) partition function calculation, using $\alpha = 1$, counting applications, using $\alpha = 0$, or computing important properties of the equilibrium energy landscape, using $\alpha = k_B T N / \varepsilon$. Here we go one step further and show that a single call with $\alpha = k_B T N / \varepsilon = \Theta(N)$, gives all these results.

We already showed one call to our magnifiable partition function algorithm $Q_\alpha(\pi)$, with $\alpha \geq k_B T N / \varepsilon$, computes the set of occupied energy levels $\mathcal{G}$ and with the number of secondary structures per level. We can compute the (usual) partition function $Q = Q_1(\pi)$ from that information, since:

$$Q_1(\pi) = \sum_{1 \leq i \leq |\mathcal{G}|} \#SSEL(g_i) \cdot e^{-g_i / k_B T}$$

where $g_i \in \mathbb{R}$ is the free energy of the i^{th} energy level that is occupied by at least one secondary structure. In other words, the partition function can be computed as a sum of products, of an energy g_i multiplied by the number of structures $\#SSEL(g_i)$ with energy g_i .

And since we established a high resolution version of our onion peeling Algorithm 7, where we have the full equilibrium landscape over $\Omega_{\max}(\pi, R)$, we can achieve all combinatorial quantities that we achieved in Theorem 34. Hence, just one magnifiable partition function call with $\alpha = k_B T N / \varepsilon$ is enough.

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A Appendix: Distinguishability complications: the partition function algorithm of Dirks et al.

As briefly mentioned earlier in Section 3.3, the Dirks et al. partition function algorithm [11] computes $\overline{Q}(\pi)$ first, then post-process $\overline{Q}(\pi)$ by dividing by $v(\pi)$ as follows:

$$\begin{aligned}
\frac{\overline{Q}(\pi)}{v(\pi)} &= \sum_{S \in \overline{\Omega}(\pi)} \frac{1}{v(\pi)} e^{-\overline{\Delta G}(S)/k_B T} \\
&= \sum_{S \in \overline{\Omega}(\pi)} \frac{1}{|\text{Orb}(S)| \cdot |\text{Stab}(S)|} e^{-\overline{\Delta G}(S)/k_B T} && \text{Using Eq. (10)} \\
&= \sum_{S \in \Omega(\pi)} e^{-\log R_{\max}(S)} e^{-\overline{\Delta G}(S)/k_B T} && \text{Since } |\text{Stab}(S)| = R_{\max}(S) \\
&= \sum_{S \in \Omega(\pi)} e^{-[\overline{\Delta G}(S) + k_B T \log R_{\max}(S)]/k_B T} \\
&= \sum_{S \in \Omega(\pi)} e^{-\Delta G(S)/k_B T} = Q(\pi)
\end{aligned}$$

This division by $v(\pi) = |\text{Orb}(S)| \cdot R_{\max}(S)$, achieves the following simultaneously:

- Cancels out overcounting by correcting the underlying state space from $\overline{\Omega}(\pi)$ to $\Omega(\pi)$. This is achieved when divided by $|\text{Orb}(S)|$, since there are $|\text{Orb}(S)|$ indistinguishable secondary structures within $\overline{\Omega}(\pi)$ that are represented as a single structure in $\Omega(\pi)$, see Section 3.4.
- Corrects the underlying free energy model from $\overline{\Delta G}(S)$ to $\Delta G(S)$, this is achieved when divided by $|\text{Stab}(S)| = R_{\max}(S)$. As a consequence the entropic penalty for rotational symmetry [1], got included, leading to $Q(\pi)$.

Dirks et al.'s algorithm benefits hugely from two things: (1) The algebraic relationship between $\overline{\Omega}(\pi)$ and $\Omega(\pi)$, see Section 3.4, and (2) The natural exponential nature of the weighted Boltzmann distribution of the partition function, otherwise the previous post-processing algebraic link will never work. We believe the simultaneous handling of Dirks et al.'s [11] algorithm of those two problems of turns out to be its main obstacle for being a magnifiable partition function algorithm.

B Appendix: Omitted proofs and algorithms

B.1 Proofs omitted from Section 5

Lemma 28. For any $(R \geq 2)$ -fold rotationally symmetric secondary structure S , for any two consecutive nests A and B with exteriors (i, j) and (k, l) respectively, any covalent bond $[a, a + 1]$ on the path of the circumference of $\text{Poly}(S, \pi)$ from A to B is an admissible R -symmetric backbone cut generator.

Proof. Given any covalent bond $b = [a, a + 1]$ on the path of the circumference of $\text{Poly}(S, \pi)$ from A to B , we claim b is an admissible R -symmetric backbone cut generator, otherwise there exists a base pair $(m, n) \in \text{Poly}(S, \pi)$ such that $b \in \llbracket m, n \rrbracket$, giving a contradiction by any of the following cases:

- $\llbracket m, n \rrbracket$ must contain either $\llbracket i, j \rrbracket$ or $\llbracket k, l \rrbracket$ or both which contradicts the fact that (i, j) and (k, l) are the exteriors of A and B respectively.
- (m, n) intersect with either (i, j) or (k, l) or both forming a pseudoknot.
- m and n bases belong to the path of the circumference of $\text{Poly}(S, \pi)$ between A and B contradicting the fact that A and B are consecutive nests.

All covalent bonds in $\bigotimes_R^b$ have the same situation because of the R -fold symmetry of S , which implies that $\bigotimes_R^b$ is an admissible R -symmetric backbone cut of S . $\square$

B.2 Proofs and algorithms omitted from Section 6

Algorithm 5 The assistant chef algorithm pseudocode (see Lemma 30), that runs in $\mathcal{O}(N^3)$ time and $\mathcal{O}(N^2)$ space with recursive calls illustrated in Figure 5. Following the same notations of [11], nicks between strands are denoted by half indices (e.g. $x + \frac{1}{2}$). The function $\eta[i + \frac{1}{2}, j + \frac{1}{2}]$ returns the number of nicks in the interval $[i + \frac{1}{2}, j + \frac{1}{2}]$. The shorthand $\eta[i + \frac{1}{2}]$ is equivalent to $\eta[i + \frac{1}{2}, i + \frac{1}{2}]$ and by convention, $\eta[i + \frac{1}{2}, i - \frac{1}{2}] = 0$.

```

1: Initialize  $Q^{\alpha,R}, Q^{\text{bp},\alpha,R}, Q^{m,\alpha,R}$  by setting all values to 0, except  $Q_{i,i-1}^{\alpha,R} = Q_{i,i}^{\alpha,R} = 1$  for all  $i = 1, \dots, N$ 
2: for  $l \leftarrow 1 \dots N/R$  do
3:   for  $i \leftarrow 1 \dots \frac{2N}{R} - l + 1$  do ▷ If  $R = 1$ , let  $i \leftarrow 1 \dots N - l + 1$ 
4:      $j = i + l - 1$  ▷  $Q^{\text{bp},\alpha,R}$  recursion equations

5:     if  $\eta[i + \frac{1}{2}, j - \frac{1}{2}] == 0$  then  $Q_{i,j}^{\text{bp},\alpha,R} = \exp\{\Delta G_{i,j}^{\text{hairpin}}/k_B T\}$  ▷ hairpin loop requires no nicks
6:     end if
7:     for  $d \leftarrow i + 1 \dots j - 2$  do ▷ loop over all possible 3'-most pairs  $(d, e)$ 
8:       for  $e \leftarrow d + 1 \dots j - 1$  do
9:         if  $\eta[i + \frac{1}{2}, d - \frac{1}{2}] == 0$  and  $\eta[e + \frac{1}{2}, j - \frac{1}{2}] == 0$  then
10:           $Q_{i,j}^{\text{bp},\alpha,R} + = \exp\{\Delta G_{i,d,e}^{\text{interior}}/k_B T\} Q_{d,e}^{\text{bp},\alpha,R}$ 
11:        end if ▷ multiloop: no nicks

12:        if  $\eta[e + \frac{1}{2}, j - \frac{1}{2}] == 0$  and  $\eta[i + \frac{1}{2}] == 0$  and  $\eta[d - \frac{1}{2}] == 0$  then
13:           $Q_{i,j}^{\text{bp},\alpha,R} + = Q_{i+1,d-1}^{m,\alpha,R} Q_{d,e}^{\text{bp},\alpha,R} \exp\{-[\Delta G_{\text{init}}^{\text{multi}} + 2\Delta G_{\text{bp}}^{\text{multi}} + (j - e - 1)\Delta G_{\text{nt}}^{\text{multi}}]/k_B T\}$ 

14:        end if
15:      end for
16:    end for
17:    for  $x \in \{i, \dots, j - 1\}$  s.t.  $\eta[x + \frac{1}{2}] = 1$  do ▷ loop over all nicks  $\in [i + \frac{1}{2}, j - \frac{1}{2}]$ 
18:      if  $(\eta[i + \frac{1}{2}] == 0$  and  $\eta[j - \frac{1}{2}] == 0)$  or  $(i == j - 1)$  or  $(x == i$  and  $\eta[j - \frac{1}{2}] == 0)$  or  $(x == j - 1$  and  $\eta[i + \frac{1}{2}] == 0)$  then
19:         $Q_{i,j}^{\text{bp},\alpha,R} + = Q_{i+1,c}^{\alpha,R} Q_{c+1,j-1}^{\alpha,R}$  ▷ exterior loops
20:      end if
21:    end for ▷  $Q^{\alpha,R}, Q^{m,\alpha,R}$  recursion equations

22:    if  $\eta[i + \frac{1}{2}, j - \frac{1}{2}] == 0$  then  $Q_{i,j}^{\alpha,R} = 1$  ▷ empty substructure
23:    else  $Q_{i,j}^{\alpha,R} = 0$  ▷ unconnected substructure
24:    end if
25:    for  $d \leftarrow i \dots j - 1$  do ▷ loop over all possible 3'-most pairs  $(d, e)$ 
26:      for  $e \leftarrow d + 1 \dots j$  do
27:        if  $\eta[e + \frac{1}{2}, j - \frac{1}{2}] == 0$  then
28:          if  $\eta[d - \frac{1}{2}] == 0$  or  $d == i$  then
29:             $Q_{i,j}^{\alpha,R} + = Q_{i,d-1}^{\alpha,R} Q_{d,e}^{\text{bp},\alpha,R}$ 
30:          end if
31:          if  $\eta[i + \frac{1}{2}, d - \frac{1}{2}] == 0$  then ▷ single base pair
32:             $Q_{i,j}^{m,\alpha,R} + = \exp\{-[\Delta G_{\text{bp}}^{\text{multi}} + (d - i + j - e)\Delta G_{\text{nt}}^{\text{multi}}]/k_B T\} Q_{d,e}^{\text{bp},\alpha,R}$ 
33:          end if
34:          if  $\eta[d - \frac{1}{2}] == 0$  then ▷ more than one base pair
35:             $Q_{i,j}^{m,\alpha,R} + = Q_{i,d-1}^{m,\alpha,R} Q_{d,e}^{\text{bp},\alpha,R} \exp\{-[\Delta G_{\text{bp}}^{\text{multi}} + (j - e)\Delta G_{\text{nt}}^{\text{multi}}]/k_B T\}$ 
36:          end if
37:        end if
38:      end for
39:    end for
40:  end for
41: end for

```

Lemma 33 (Master Chef). Given any multi-stranded system of strands $\mathcal{S}$ with strand ordering π , magnification factor α , and valid symmetry degree $R \geq 2$ such that $R|v(\pi)$:

- if $R > 2$, Eq. (39) computes $\overline{Q}_\alpha(\pi, R)$ correctly. Formally speaking:

$$\overline{Z}_S^{\alpha, R} = \sum_{S \in \overline{\Omega}(\pi, R)} e^{-\alpha \overline{\Delta G}(S)/k_B T} = \overline{Q}_\alpha(\pi, R).$$

- if $R = 2$, Eq. (40) computes $\overline{Q}_\alpha(\pi, 2)$ correctly. Formally speaking:

$$\overline{Z}_S^{\alpha, 2} = \sum_{S \in \overline{\Omega}(\pi, 2)} e^{-\alpha \overline{\Delta G}(S)/k_B T} = \overline{Q}_\alpha(\pi, 2).$$

Furthermore, Eqs. (39) and (40) are implemented in the master chef Algorithm 6, that has asymptotic complexity of $\mathcal{O}(N^2)$ time, and $\mathcal{O}(1)$ space.

Proof. Let's start with Eq. (39) for the case when $R > 2$. It's a double summation of two terms:

- $t_1 = \mathbb{1}_{x,i} Q_{i,j}^{\text{bp}, \alpha, R} e^{-\alpha [\Delta G_{\text{init}}^{\text{multi}} + R \cdot \Delta G_{\text{bp}}^{\text{multi}} + R \cdot (i-x) \Delta G_{\text{nt}}^{\text{multi}} + \beta]/k_B T}$, that captures the case when there is exactly one nest will be formed in the template slice Δ_R^b .
- $t_2 = Q_{x,i-1}^{m, \alpha, R} \mathbb{1}_i Q_{i,j}^{\text{bp}, \alpha, R} e^{-\alpha [\Delta G_{\text{init}}^{\text{multi}} + R \cdot \Delta G_{\text{bp}}^{\text{multi}} + \beta]/k_B T}$, that captures the only other case when there is more than one nest will be formed in Δ_R^b .

Let's simplify the first term t_1 first, see Figure 7 (a). By substituting $Q_{i,j}^{\text{bp}, \alpha, R}$ using Eq. (36), and reordering:

$$t_1 = e^{-\alpha [\Delta G_{\text{init}}^{\text{multi}} + R \cdot \Delta G_{\text{bp}}^{\text{multi}} + R \cdot (i-x) \Delta G_{\text{nt}}^{\text{multi}} + \beta]/k_B T} \mathbb{1}_{x,i} \sum_{r \in \Omega_{i,j}^{\text{bp}}} e^{-\alpha R \Delta G(r)/k_B T} \quad (51)$$

If there is any nick in the interval $[x, i]$, where $x = \mathcal{B}_1(\Delta_R^b)$, then the substructure that will be within the template slice Δ_R^b is disconnected, and hence neglected in the partition function calculation by multiplication by $\mathbb{1}_{x,i} = 0$. On the other hand, if $\mathbb{1}_{x,i} = 1$, we end up concatenating the empty connected substructure over the interval $[x, i]$, with any connected substructure $r \in \Omega_{i,j}^{\text{bp}}$. This leads to form a connected slice. After repeating that formed slice R times, joining each to the next by a covalent bond where our R -symmetric cut was initiated, we form an R -fold rotationally symmetric secondary structure S .

Since the base pair (i, j) , where $j = \mathcal{B}_2(\Delta_R^b)$, is chosen such that it crosses the second border of the prime template's slice, hence it satisfies the requirement of $\mathcal{P}$ in our main criterion, see Eq. (38). So, $S \in \mathcal{P}(\Delta_R^b, (i, j), \#n=1)$, see Note 32. Now we can replace the dummy variable r , in Eq. (51), with the actual slice $\Delta_R^{S,b}$ of S since they are equivalent.

$$\begin{aligned} t_1 &= e^{-\alpha [\Delta G_{\text{init}}^{\text{multi}} + R \cdot \Delta G_{\text{bp}}^{\text{multi}} + R \cdot (i-x) \Delta G_{\text{nt}}^{\text{multi}} + \beta]/k_B T} \sum_{S \in \mathcal{P}(\Delta_R^b, (i, j), \#n=1)} e^{-\alpha R \Delta G(\Delta_R^{S,b})/k_B T} \\ &= \sum_{S \in \mathcal{P}(\Delta_R^b, (i, j), \#n=1)} e^{-\alpha [R \Delta G(\Delta_R^{S,b}) + \Delta G_{\text{init}}^{\text{multi}} + R [\Delta G_{\text{bp}}^{\text{multi}} + (i-x) \Delta G_{\text{nt}}^{\text{multi}}] + \beta]/k_B T} \end{aligned} \quad (52)$$

Since $R > 2$, Remark 29 shows that the central loop $\mathcal{C}(S)$ must be a multiloop. Since there is exactly one nest per slice $\Delta_R^{S,b}$, therefore one exterior base pair per slice that borders the interior

of $\mathcal{C}(S)$. Each slice contributes to the free energy of $\mathcal{C}(S)$ with exactly $\Delta G_{\text{bp}}^{\text{multi}} + (i - x)\Delta G_{\text{nt}}^{\text{multi}}$. The free energy of the multiloop $\Delta G(\mathcal{C}(S)) = \Delta G_{\text{init}}^{\text{multi}} + R[\Delta G_{\text{bp}}^{\text{multi}} + (i - x)\Delta G_{\text{nt}}^{\text{multi}}]$. Substitute $\Delta G(\mathcal{C}(S))$ and β in Eq. (52), we get the following:

$$\begin{aligned} t_1 &= \sum_{S \in \mathcal{P}(\Delta_R^b(i, j), \#n=1)} e^{-\alpha [R\Delta G(\Delta_R^{S, b}) + \Delta G(\mathcal{C}(S)) + (c-1)\Delta G^{\text{assoc}}] / k_B T} \\ &= \sum_{S \in \mathcal{P}(\Delta_R^b(i, j), \#n=1)} e^{-\alpha \overline{\Delta G}(S) / k_B T}, \end{aligned} \quad \text{using Eq. (28)} \quad (53)$$

With respect to the second term t_2 , that captures the case when there is more than one nest will be formed in Δ_R^b , see Figure 7 (a). By substituting $Q_{i,j}^{m, \alpha, R}$ and $Q_{i,j}^{\text{bp}, \alpha, R}$ using Eqs. (37) and (36), and reordering we get the following:

$$\begin{aligned} t_2 &= \mathbb{1}_i \left[\sum_{r \in \Omega_{x, i-1}} e^{-\alpha R [\Delta G(r) + \# \text{bp}(r) \cdot \Delta G_{\text{bp}}^{\text{multi}} + \# \text{nt}(r) \cdot \Delta G_{\text{nt}}^{\text{multi}}] / k_B T} \right] \left[\sum_{r' \in \Omega_{i,j}^{\text{bp}}} e^{-\alpha R \Delta G(r') / k_B T} \right] \\ &\quad \cdot e^{-\alpha [\Delta G_{\text{init}}^{\text{multi}} + R \cdot \Delta G_{\text{bp}}^{\text{multi}} + \beta] / k_B T} \end{aligned} \quad (54)$$

If $\mathbb{1}_i = 0$, then there is a nick between bases $i - 1$ and i . therefore the substructure that will be formed within the template slice Δ_R^b is disconnected, and hence neglected in the partition function calculation using multiplication by $\mathbb{1}_i = 0$. On the other hand, if $\mathbb{1}_i = 1$, we end up concatenating two connected substructures $r \in \Omega_{x, i-1}$ and $r' \in \Omega_{i,j}^{\text{bp}}$ using the covalent bond $[i - 1, i]$. Concatenating r and r' results in a new connected substructure $t \in \Omega_{x,j}$ such that $\Delta G(t) = \Delta G(r) + \Delta G(r')$. Since $R > 2$, t contributes to the central multiloop with $\# \text{nt}(t) = \# \text{nt}(r)$ free bases²⁴, and $\# \text{bp}(t) = \# \text{bp}(r) + 1$ base pairs that border the interior of the central multiloop. Note that the extra base pair comes from the nest with exterior base pair (i, j) from substructure r' . Reflecting these changes in Eq. (54), we get the following:

$$\begin{aligned} t_2 &= \sum_{r \in \Omega_{x, i-1}} \sum_{r' \in \Omega_{i,j}^{\text{bp}}} e^{-\alpha [R(\Delta G(r) + \Delta G(r')) + \Delta G_{\text{init}}^{\text{multi}} + R(\# \text{bp}(r) + 1) \Delta G_{\text{bp}}^{\text{multi}} + R \cdot \# \text{nt}(r) \Delta G_{\text{nt}}^{\text{multi}} + \beta] / k_B T} \\ &= \sum_{t \in \Omega_{x,j}} e^{-\alpha [R \cdot \Delta G(t) + \Delta G_{\text{init}}^{\text{multi}} + R \cdot \# \text{bp}(t) \cdot \Delta G_{\text{bp}}^{\text{multi}} + R \cdot \# \text{nt}(t) \cdot \Delta G_{\text{nt}}^{\text{multi}} + \beta] / k_B T} \end{aligned} \quad (55)$$

Since $t \in \Omega_{x,j}$ is a connected substructure over the interval $[x, j]$, where $x = \mathcal{B}_1(\Delta_R^b)$, $j = \mathcal{B}_2(\Delta_R^b)$, in other words the template slice Δ_R^b . Therefore t is a connected slice. After repeating that formed slice R times, joining each to the next by a covalent bond where our cut was done, we form an R -fold rotationally symmetric secondary structure S . And since the base pair (i, j) is chosen such that it crosses the second border of the prime template's slice, so it satisfies the requirement of $\mathcal{P}$ in our main criterion, see Eq. (38). So, $S \in \mathcal{P}(\Delta_R^b(i, j), \#n > 1)$, see Note 32. We then can replace the dummy variable t , in Eq. (55), with the actual slice $\Delta_R^{S, b}$ of S since they are equivalent.

$$t_2 = \sum_{S \in \mathcal{P}(\Delta_R^b(i, j), \#n > 1)} e^{-\alpha [R \cdot \Delta G(\Delta_R^{S, b}) + \Delta G_{\text{init}}^{\text{multi}} + R \cdot \# \text{bp}(\Delta_R^{S, b}) \cdot \Delta G_{\text{bp}}^{\text{multi}} + R \cdot \# \text{nt}(\Delta_R^{S, b}) \cdot \Delta G_{\text{nt}}^{\text{multi}} + \beta] / k_B T}$$

²⁴Note that, $r' \in \Omega_{i,j}^{\text{bp}}$ contributes with zero free bases to the central multiloop, since r' is just one nest with exterior base pair (i, j) .

Since S has a central multiloop $\mathcal{C}(S)$, the free energy of $\mathcal{C}(S)$ is as follows: $\Delta G(\mathcal{C}(S)) = \Delta G_{\text{init}}^{\text{multi}} + R[\# \text{bp}(\blacktriangle_R^{S,b}) \cdot \Delta G_{\text{bp}}^{\text{multi}} + \# \text{nt}(\blacktriangle_R^{S,b}) \cdot \Delta G_{\text{nt}}^{\text{multi}}]$. Substitute $\Delta G(\mathcal{C}(S))$ and β in Eq. (55), we obtain the following:

$$\begin{aligned} t_2 &= \sum_{S \in \mathcal{P}(\blacktriangle_R^b, (i,j), \#n > 1)} e^{-\alpha [R\Delta G(\blacktriangle_R^{S,b}) + \Delta G(\mathcal{C}(S)) + (c-1)\Delta G^{\text{assoc}}] / k_B T} \\ &= \sum_{S \in \mathcal{P}(\blacktriangle_R^b, (i,j), \#n > 1)} e^{-\alpha \overline{\Delta G}(S) / k_B T}, \end{aligned} \quad \text{using Eq. (28)} \quad (56)$$

Now, by substituting t_1 and t_2 in Eq. (39), we get:

$$\begin{aligned} \overline{Z}_S^{\alpha, R} &= \sum_{\blacktriangle_R^b \in \mathcal{T}_R} \sum_{(i,j)} \left[\sum_{S \in \mathcal{P}(\blacktriangle_R^b, (i,j), \#n=1)} e^{-\alpha \overline{\Delta G}(S) / k_B T} + \sum_{S \in \mathcal{P}(\blacktriangle_R^b, (i,j), \#n > 1)} e^{-\alpha \overline{\Delta G}(S) / k_B T} \right] \\ &= \sum_{\blacktriangle_R^b \in \mathcal{T}_R} \sum_{(i,j)} \sum_{S \in \mathcal{P}(\blacktriangle_R^b, (i,j))} e^{-\alpha \overline{\Delta G}(S) / k_B T} \quad \text{using Note 32} \\ &= \sum_{S \in \overline{\Omega}(\pi, R)} e^{-\alpha \overline{\Delta G}(S) / k_B T} \quad \text{by Eq. (38) in our criterion, see Lemma 31} \\ &= \overline{Q}_\alpha(\pi, R) \quad \text{by definition} \end{aligned}$$

This proves our first case when $R > 2$.

When $R = 2$, see Eq. (40), the proof goes through the same lines except for the first case when there is only one nest per slice. In this case, the central loop $\mathcal{C}(S)$ of the formed secondary structure S is internal loop or stack. $\mathcal{C}(S)$ is determined by its four bases i, j, i' , and j' , where $i' = i + \frac{N}{2}$ and $j' = (j + \frac{N}{2}) \bmod N$. The free energy of $\mathcal{C}(S)$ is $\Delta G(\mathcal{C}(S)) = \Delta G_{i,j,i',j'}^{\text{interior}}$. In conclusion this shows that $\overline{Z}_S^{\alpha, 2} = \sum_{S \in \overline{\Omega}(\pi, 2)} e^{-\alpha \overline{\Delta G}(S) / k_B T} = \overline{Q}_\alpha(\pi, 2)$.

With respect to time and space complexity, each of Eqs. (39) and (40) has double summation. The first summation ranges over $|\mathcal{T}_R| = \frac{N-c}{R} = \mathcal{O}(N)$, see Section 5. And in the second summation, base i of base pair (i, j) ranges²¹ as follows: $\mathcal{B}_1(\blacktriangle_R^b) \leq i < \mathcal{B}_2(\blacktriangle_R^f)$, since $\mathcal{B}_2(\blacktriangle_R^f) - \mathcal{B}_1(\blacktriangle_R^b) < \frac{N}{R} = \mathcal{O}(N)$. Hence we have $\mathcal{O}(N^2)$ iterations, each iteration uses the already pre-computed values from the Assistant Chef Algorithm 5, and hence each is done in $\mathcal{O}(1)$ time and space. In total our Master Chef Algorithm 6 runs in $\mathcal{O}(N^2)$ time, and $\mathcal{O}(1)$ space, proving our claim. $\square$

Algorithm 6 The master chef algorithm pseudocode (see Lemma 33), that runs in $\mathcal{O}(N^2)$ time and $\mathcal{O}(1)$ space with recursive calls illustrated in Figure 7. The indicator variables $\mathbb{1}_{a,b}$ and $\mathbb{1}_a$ are defined in Eqs. (41) and (42) respectively.

```

1: Initialize  $\overline{Z}_S^{\alpha,R} = 0$ 
2: Set  $\beta = (c-1)\Delta G^{\text{assoc}}$ 
3: for  $a \leftarrow 1 \dots \frac{N}{R} - 1$  do
4:    $b = [a, a+1]$  ▷ The covalent bond that will be used as backbone cut generator
5:    $x = a + 1$  ▷  $x = \mathcal{B}_1(\Delta_R^b) = a + 1$ 
6:    $j = a + \frac{N}{R}$  ▷  $j = \mathcal{B}_2(\Delta_R^b) = a + \frac{N}{R}$ 

7:   for  $i \leftarrow x \dots 1 + \frac{N}{R}$  do ▷  $\mathcal{B}_1(\Delta_R^b) \leq i < \mathcal{B}_2(\Delta_R^f)$ 

8:     if  $\mathbb{1}_{x,i} == 1$  then ▷ Case 1: Only one nest per slice
9:       if  $R == 2$  then
10:         $\overline{Z}_S^{\alpha,R} += Q_{i,j}^{\text{bp},\alpha,R} \exp \left\{ -\alpha [\Delta G_{i,j,i',j'}^{\text{interior}} + \beta] / k_B T \right\}$ 
11:      end if

12:      if  $R > 2$  then
13:         $\overline{Z}_S^{\alpha,R} += Q_{i,j}^{\text{bp},\alpha,R} \exp \left\{ -\alpha [\Delta G_{\text{init}}^{\text{multi}} + R \cdot \Delta G_{\text{bp}}^{\text{multi}} \right.$ 
+  $(i-x)\Delta G_{\text{nt}}^{\text{multi}} + \beta] / k_B T \}$ 
14:      end if
15:    end if

16:    if  $\mathbb{1}_i == 1$  then ▷ Case 2: More than one nest per slice
17:       $\overline{Z}_S^{\alpha,R} += Q_{x,i-1}^{m,\alpha,R} Q_{i,j}^{\text{bp},\alpha,R} \exp \left\{ -\alpha [\Delta G_{\text{init}}^{\text{multi}} + R \cdot \Delta G_{\text{bp}}^{\text{multi}} + \beta] / k_B T \right\}$ 
18:    end if
19:  end for
20: end for
21: Return  $\overline{Z}_S^{\alpha,R}$ 

```

B.3 Proofs and algorithms omitted from Section 7

Theorem 3. There is an $\mathcal{O}(N^3(c-1)!)$ time and $\mathcal{O}(N^2)$ space algorithm for the #SSEL (density of states, or DOS) nearest neighbour unpseudoknotted secondary structure prediction problem for a set of $c = \mathcal{O}(1)$ DNA/RNA strands of total length N bases.

Proof. For any strand ordering $\pi \in \Pi$, let $\mathcal{G}_\pi = \{\Delta G(S) \mid S \in \Omega(\pi)\} = \{g_1 < g_2 < \dots < g_{|\mathcal{G}_\pi|}\}$ denote the set of **occupied** energy levels of system $\mathcal{S}$. Let $\#\text{SSEL}(g_i, \pi)$ denote the number of secondary structures $S \in \Omega(\pi)$ such that $\Delta G(S) = g_i$. Note that for any energy level $g_i \in \mathcal{G}$, $\#\text{SSEL}(g_i, \pi) > 0$, and $g_1 = \text{MFE}(\pi)$. In Algorithm 7, we claim the following about the returned arrays $\mathcal{L}$ and $\mathcal{N}$:

- For every $1 \leq i \leq |\mathcal{G}_\pi|$, $\mathcal{L}[i] = g_i$ and $\mathcal{N}[i] = \#\text{SSEL}(g_i, \pi)$.

We prove this claim by mathematical induction on arrays index i .

In Algorithm 7, note that $\mathcal{R}$ is a loop variant that gets updated at the end of every loop iteration. Let $\mathcal{R}_j$ denote the value of $\mathcal{R}$ after the j^{th} iteration, the value of $\mathcal{R}$ is updated according to the

following rule: $\mathcal{R}_j = \mathcal{R}_{j-1} - \mathcal{N}[j] \cdot e^{-\alpha \mathcal{L}[j]/k_B T}$, see line 8. Note that $\mathcal{R}_j$ is used during iteration $j + 1$, and the initial value of $\mathcal{R}$ is $\mathcal{R}_0 = Q_\alpha(\pi)$, see line 3.

- **Base case:** When $i = 1$ (just after the loop first iteration), since $\mathcal{R}_0 = Q_\alpha(\pi)$, Theorem 2 guarantees that $\mathcal{L}[1] = \text{MFE}(\pi) = g_1$, and Theorem 41 guarantees that $\mathcal{N}[1] = \#\text{SSEL}(\text{MFE}(\pi), \pi) = \#\text{SSEL}(g_1)$, proving the base case. And $\mathcal{R}_0$ gets updated to $\mathcal{R}_1 = Q_\alpha(\pi) - \mathcal{N}[1] \cdot e^{-\alpha \mathcal{L}[1]/k_B T}$ after the end of the first iteration.
- **Induction Hypothesis:** For all $1 \leq i < k$, assume $\mathcal{L}[i] = g_i$, and $\mathcal{N}[i] = \#\text{SSEL}(g_i, \pi)$.
- **Induction step:** When $i = k$ (just after the loop k^{th} iteration), in Algorithm 7, lines 6 and 7, $\mathcal{L}[k] = \lceil -k_B T \log \mathcal{R}_{k-1}/(\alpha \cdot \varepsilon) \rceil \cdot \varepsilon$, and $\mathcal{N}[k] = \lfloor \mathcal{R}_{k-1}/e^{-\alpha \mathcal{L}[k]/k_B T} \rfloor$. Applying the induction hypothesis on $\mathcal{R}_{k-1}$:

$$\begin{aligned}
\mathcal{R}_{k-1} &= \mathcal{R}_{k-2} - \mathcal{N}[k-1] \cdot e^{-\alpha \mathcal{L}[k-1]/k_B T} \\
&= \mathcal{R}_{k-2} - \#\text{SSEL}(g_{k-1}, \pi) \cdot e^{-\alpha \cdot g_{k-1}/k_B T} \\
&= Q_\alpha(\pi) - \sum_{1 \leq i \leq k-1} \#\text{SSEL}(g_i, \pi) \cdot e^{-\alpha \cdot g_i/k_B T} \\
&= \sum_{1 \leq i \leq |\mathcal{G}_\pi|} \#\text{SSEL}(g_i, \pi) \cdot e^{-\alpha \cdot g_i/k_B T} - \sum_{1 \leq i \leq k-1} \#\text{SSEL}(g_i, \pi) \cdot e^{-\alpha \cdot g_i/k_B T} \\
&= \sum_{k \leq i \leq |\mathcal{G}|} \#\text{SSEL}(g_i, \pi) \cdot e^{-\alpha \cdot g_i/k_B T} \\
&= \sum_{S \in B} e^{-\alpha \cdot \Delta G(S)/k_B T} \quad \text{such that } B = \{S \in \Omega(\pi) \mid \Delta G(S) \geq g_k\}
\end{aligned}$$

In other words, $\mathcal{R}_{k-1}$ is the magnified partition function over B . Hence we can think of $\mathcal{R}_{k-1}$ as the magnified partition function of a new system $\mathcal{S}'$, where B is the set of secondary structures under interest of $\mathcal{S}'$. Since $|B| < |\Omega(\pi)|$, the hierarchical dominance Lemma 39 applies on $\mathcal{S}'$.

Hence, we now can apply the $\text{MFE}(\pi)$ algorithm, see Theorem 2, on $\mathcal{S}'$, and it guarantees that $\mathcal{L}[k] = g_k$ (The minimum free energy level over B). And apply the $\#\text{SSEL}(\pi)$ algorithm, see Theorem 41, also guarantees that $\mathcal{N}[k] = \#\text{SSEL}(g_k, \pi)$, proving the induction step.

Hence, Algorithm 7 computes the sorted set of occupied levels $\mathcal{G}_\pi$, along with the number of secondary structures per level $\#\text{SSEL}(L, \pi)$ for all energy levels with respect to any strand ordering $\pi \in \Pi$. Since $|\mathcal{G}_\pi| = \mathcal{O}(N)$, see Lemma 37, the asymptotic time and space complexity of Algorithm 7 is $\mathcal{O}(N^3)$ and $\mathcal{O}(N^2)$ respectively, as it involves exactly one call to our magnifiable partition function $Q_\alpha(\pi)$.

Since $\#\text{SSEL}(L) = \sum_{\pi \in \Pi} \#\text{SSEL}(L, \pi)$, where $|\Pi| \leq (c-1)!$. Since, different circular permutations π may have overlapping energies, a simple merge step (i.e. takes $\mathcal{O}(N)$ time and space) after each run of Algorithm 7 on each $\pi \in \Pi$, we compute the sorted set of occupied levels $\mathcal{G}$, along with the number of secondary structures per level $\#\text{SSEL}(L)$ for all energy levels of any system $\mathcal{S}$. The asymptotic time and space complexity of our algorithm are $\mathcal{O}(N^3(c-1)!)$ and $\mathcal{O}(N^2)$ respectively. $\square$

Algorithm 7 The Onion Peeling Algorithm for the nearest neighbour energy model, that takes as input: $c = \mathcal{O}(1)$ strands with total number of bases (length) N , and strand ordering π . Runs in $\mathcal{O}(N^3)$ time and $\mathcal{O}(N^2)$ space. Proof of correctness: Theorem 3.

```

1: Let  $\mathcal{L}$  and  $\mathcal{N}$  be empty arrays                                ▷ Initialize  $\mathcal{L}$  to be an empty array
2: Let  $i = 0$                                                         ▷ Array counter used for our induction
3: Let  $\mathcal{R} = Q_\alpha(\pi)$  for  $\alpha = k_B T N / \varepsilon$                     ▷ Using Algorithm 3
4: repeat
5:    $i = i + 1$ 

6:    $\mathcal{L}[i] = \left\lceil -k_B T \log(\mathcal{R}) / (\alpha \cdot \varepsilon) \right\rceil \cdot \varepsilon$                                 ▷ See Theorem 2
7:    $\mathcal{N}[i] = \left\lfloor \mathcal{R} / e^{-\alpha \cdot \mathcal{L}[i] / k_B T} \right\rfloor$                                 ▷ See Theorem 41
8:    $\mathcal{R} = \mathcal{R} - \mathcal{N}[i] \cdot e^{-\alpha \mathcal{L}[i] / k_B T}$ 
9: until  $\mathcal{R} > 0$ 
10: Return  $\mathcal{L}, \mathcal{N}$ 

```
